

**280.B37 Strategien zur nachhaltigen und resilienten
Raumentwicklung - 'Housing Markets and regional
effects of housing shortages'**

FINAL REPORT

Dynamics of the Housing Market in Salzburg and its suburbs

2025S

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INTRODUCTION

Geographical Context and Framework Conditions of the City of Salzburg and Its Surroundings

The city of Salzburg is the capital of the federal state of the same name in western Austria and represents a significant urban, economic, and cultural hub within the region. Geographically, Salzburg is located at the northern edge of the Alps, in the transitional zone between the Alpine foothills and the Northern Limestone Alps. The city is traversed by the Salzach River, which runs through the urban area from south to north. Situated in close proximity to the Bavarian border, Salzburg is strategically embedded in a supra-regional network of transportation and economic relations, with connections to Munich, Linz, and Vienna, as well as southwards into the Alpine region.

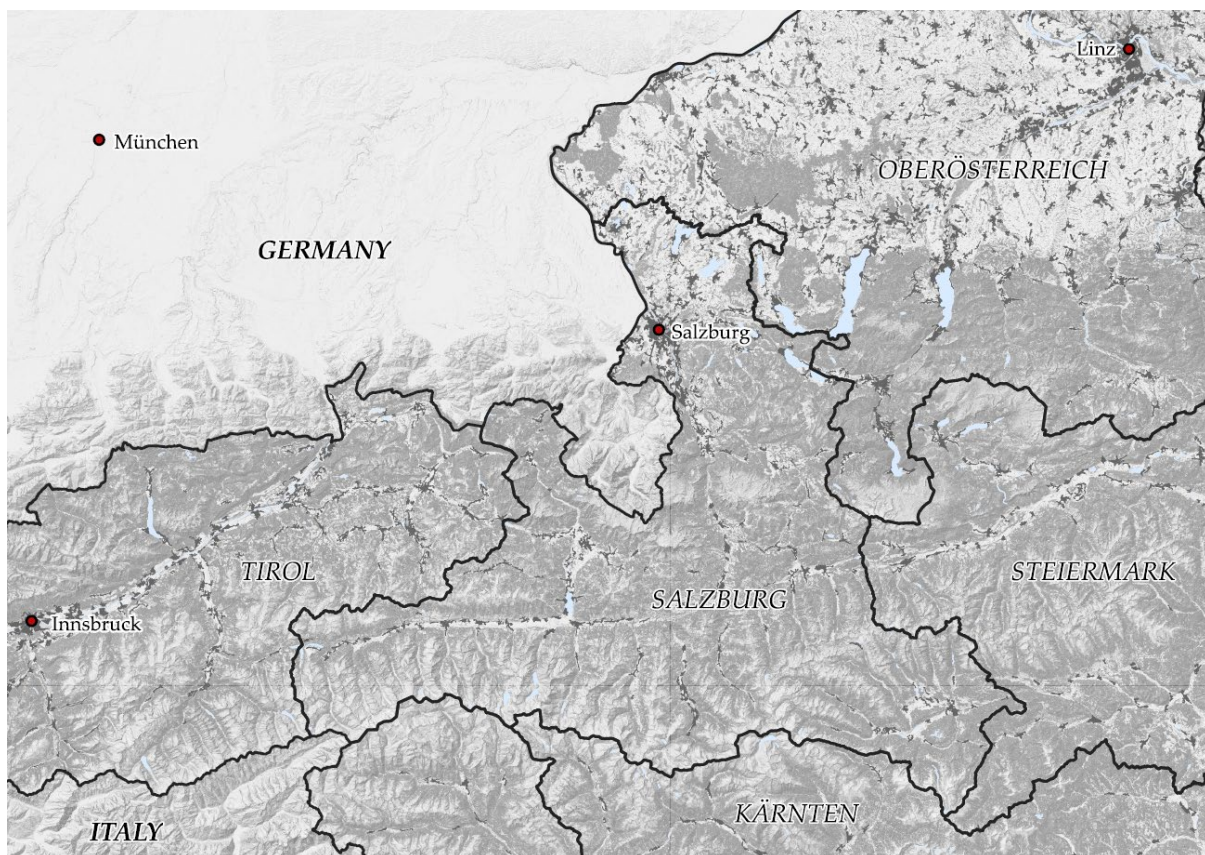


Figure 1: Location of Salzburg (data.gv.at, own representation)

The city itself is topographically shaped by several hills – including the Mönchsberg, the Kapuzinerberg, and the Festungsberg – which have historically influenced the development and spatial organization of the urban structure. The surrounding region of Salzburg is characterized by a diverse landscape, comprising lakes, forests, and agricultural areas. The districts of Flachgau to the north and Tennengau to the south constitute the immediate hinterland and are closely interlinked with the urban development of Salzburg.

A defining feature of the city is its historic Old Town, which has been designated as a UNESCO World Heritage Site. This urban heritage represents both cultural capital and regulatory

constraints in terms of urban expansion and structural modifications within the city center. Moreover, Salzburg is internationally recognized as the birthplace of Wolfgang Amadeus Mozart and as the venue of the annual Salzburg Festival, making it a focal point of cultural tourism. The influx of tourists, particularly during peak seasons, leads to temporary spatial densification and indirectly affects the housing market.

The region is further distinguished by its scenic natural landscape, particularly the nearby lake district of the Salzkammergut and the surrounding Alpine foothills. This geographical setting contributes to the city's high residential attractiveness, while simultaneously intensifying the demand for housing.

In terms of infrastructure, Salzburg is well connected via the Westbahn railway line and the motorways A1 (Westautobahn) and A10 (Tauernautobahn), facilitating both commuter flows from the surrounding areas and tourist access. Nevertheless, Salzburg is spatially constrained, with limited areas for urban expansion. This spatial pressure is further intensified by natural barriers, protected heritage zones, and land-use planning regulations.

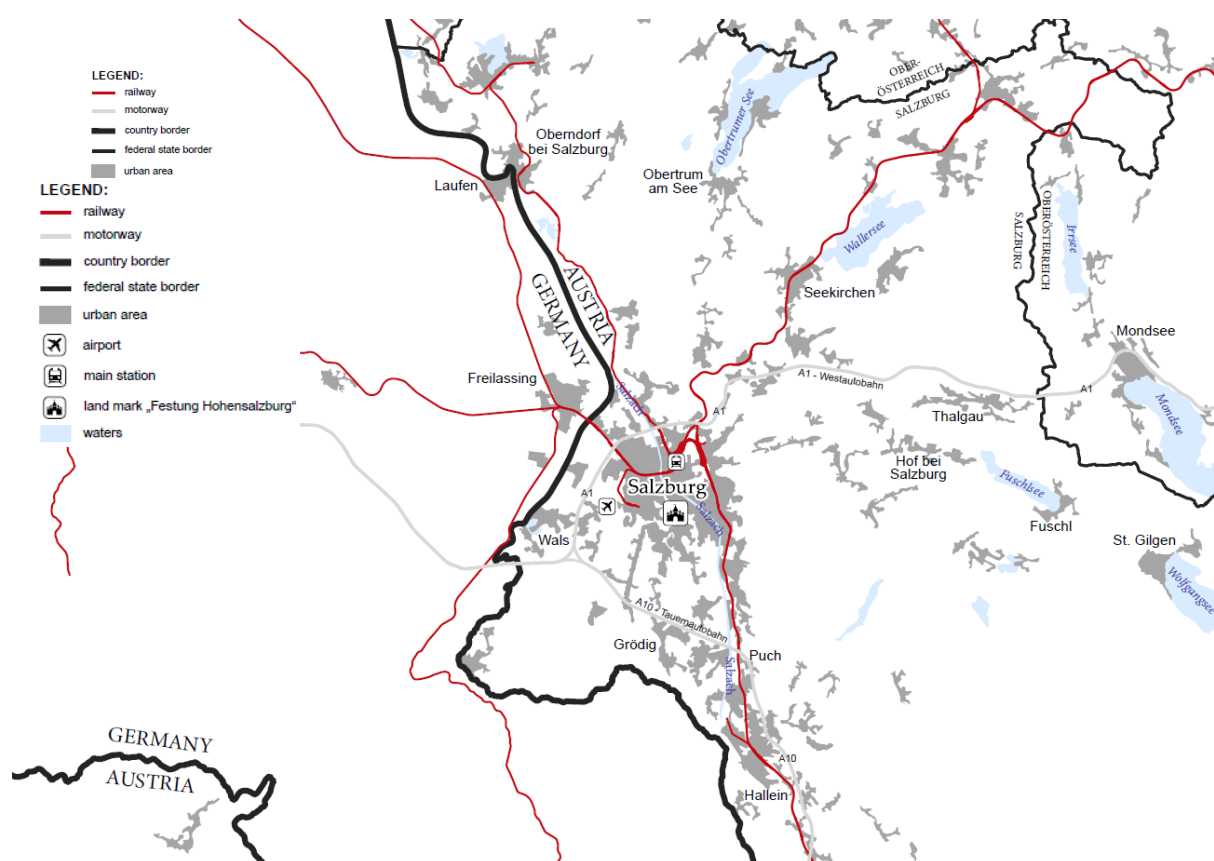


Figure 2: Structural Map of Salzburg (data.gv.at, own representation)

These specific structural conditions – including geographical limitations, strong tourism appeal, cultural significance, and restricted expansion capacities – have a direct impact on the housing market. Salzburg has long exhibited a high demand for residential space, which is met by a relatively limited supply. This imbalance has led to persistently high real estate prices and rental costs. Urban densification, increasing commuter numbers, and demographic changes pose additional challenges to housing provision.

The subsequent chapters of this report will provide a detailed analysis of these influencing factors and examine their relationship to empirical developments in Salzburg's urban and regional housing market.

Definition of the research area for further analysis

For the subsequent analytical work, it was necessary to filter the available and relevant data in order to ensure comparability. This required the establishment of a fixed delineation of the Salzburg metropolitan area. For this purpose, we followed the regional subdivision defined in the annual housing market reports "Wohnen in Salzburg Stadt und Umgebung" (Team Rauscher Immobilien GmbH, 2024). Based on this classification, in addition to the region of Salzburg City, five further surrounding subregions are defined. Figure 3 illustrates this spatial division. The white boundary lines represent individual municipalities, while the smaller grey areas indicate cadastral municipalities (Katastralgemeinden). Cadastral municipalities constitute the smallest administrative unit in Austria and were primarily used as the spatial basis for data analysis (for further methodological details, see chapter *Spatial Regression Analysis*).

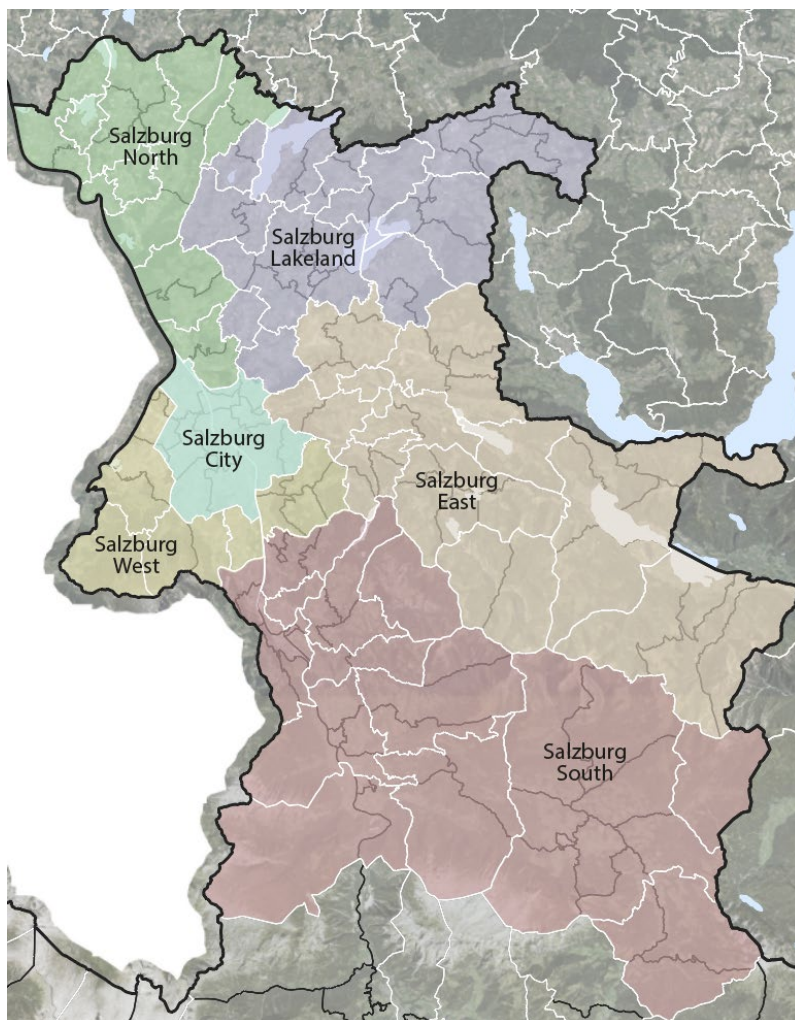


Figure 3: Definition of the Research Area and Regional Structuring of the Greater Salzburg Area (source: data.gv.at, own representation)

The data set

Housing data is working closely with the data developed in the ESPON project 'House4All'. The data set includes the location of rental and sale apartment and housing in the Salzburg city and surrounding regions, and provides detailed information about housing characteristics, such as listing type (sale, rent), housing type (apartment, house), price, duration days and size (m²). Based on these data, we can conclude the basic understanding about individual elements of the housing market.

Socio-demographic data are mainly from The Census Grid 2021, which is conducted by EUROSTAT. Its present key topics form the 2021 Census on an EU-wide 1 km² grid. The Datasets are available for the 13 variables as defined in EU regulation 1799/2018 for:

- **sex** (male and female)
- **age** (under 15 years, 15-64, 65 years and over)
- **current activity status** (number of employed persons), voluntary bases
- **country / place of birth** (place of birth in reporting country, place of birth in other EU country, place of birth elsewhere)
- **place of usual residence** (total population)
- **place of usual residence one year prior to the census**

Public services data such as coverage percentage of the population of public transportation systems used in further analysis is conducted by Openstreetmap.

The boundaries of Katastralgemeinde are from Kataster Grafik SHP Stichtag 01.04.2025 conducted by Bundesamt für Eich- und Vermessungswesen.

RESEARCH

What is our goal in this research?

This part of the report investigates the dynamics of the Salzburg housing market, focusing on rental and ownership trends, the influence of financialization, and the impact of short-term rentals (STRs), particularly Airbnb. It also includes an analysis of spatial clustering to explore the internal segmentation and micro-market structure of the region.

Main research questions:

1. *What factors have contributed to Salzburg becoming the most expensive rental region in Austria?*
2. *What are the factors influencing the dynamics of the housing market in Salzburg?*
3. *What are the spatial patterns and segmentation within the Salzburg housing market?*
4. *How has financialization influenced investment behavior and housing development?*

In light of these research questions, a comprehensive literature review and both qualitative and quantitative analyses were conducted. The findings identified two major global factors influencing housing dynamics and interrelations: short-term rentals and the COVID-19 pandemic. With these two major factors, Salzburg's housing market reflects both national and global trends of urban affordability crises, shaped by demand-side pressures (e.g., tourism, capital influx) and constrained supply due to limited land and strict planning laws (Smigiel, 2023).

Overview and development of the housing market in Salzburg

Market Trends and Dynamics (2019-2025)

Salzburg is the most expensive federal state in Austria with its average rent of €11.90 per m² in 2024, ahead of Tirol (€11.50) and Vorarlberg (€11.30) while the average rent in peripheral states like Carinthia (€7.70) and Burgenland (€7.40) remains much lower (Statistik Austria, 2024). Regina Fuchs of Statistik Austria says, "Rental costs have risen throughout Austria in recent years, but Salzburg stands out in particular." and a recent study shows that Salzburg residents are left with 5% less residual income than the national average after covering housing costs. This number jumps up to 15% when comparing Salzburg to Lower or Upper Austria (WIFO, 2022).

This is attributed to a combination of factors: high land and construction costs, limited social housing, high demand, restricted development areas. The result is Salzburg losing 26,000 of its population, experiencing Austria's largest net internal out-migration from 2001 to 2021.

By 2023, stricter lending policies introduced by the central bank and ECB along with the increase in interest rates have begun to cool the market. House and flat prices fell by approximately 2.6% across the nation, which marks the first drop in years. 2023 also saw

apartment prices decreasing by 4.3% in Salzburg city but rents have continued to rise due to inflation and lack of supply (Statistik Austria, 2023).

Mid-sized and averagely make up most of the demand in Salzburg, especially among middle-class households. Smaller units were also exchanged at a high rate, but the high €/m² and financing conditions played a limiting role. Larger and/or luxury properties, such as villas, remained on the market for longer when compared to these properties.

Clustering

Clustering analyses offer a powerful tool to understand the spatial structure of Salzburg's housing market. Grouping neighborhoods or municipalities based on variables such as price per square meter, growth rate, and proximity to the city can reveal meaningful clusters that reflect the region's internal differentiation. For instance, Salzburg's city center and inner suburbs may form one cluster characterized by high prices and tight supply, while tourist-heavy Alpine towns like Zell am See and Saalbach — driven by second home and leisure demand — could represent another. In contrast, peripheral rural communities with lower prices and distinct demand factors would likely form a separate group. Indeed, the "Gau" districts of Salzburg display unique housing profiles; for example, Pinzgau/Pongau are marked by a high concentration of holiday properties, whereas Flachgau/Tennengau are primarily commuter towns. These patterns can be quantitatively validated using unsupervised learning algorithms such as k-means or hierarchical clustering (Helbich et al, 2013). A student-led project at the University of Salzburg, analyzing around 20,000 real estate listings from 2006 to 2016, clearly demonstrated price disparities between the city and its outskirts, reinforcing the segmented nature of the housing market. Although machine learning clustering techniques were not explicitly applied in that study, it highlighted strong spatial segmentation — for instance, the fact that only one-third of university students live within the city itself suggests cost-driven spatial sorting.

Short Term Rentals

Studies have highlighted the growing influence of transnational capital as a driver behind the increase in the prominence of short-term rentals (STRs) and gentrification. STRs are increasingly viewed as part of the "fifth wave of gentrification" in Salzburg and are characterized as the urban materialization of financialized capitalism (Aalbers, 2019). The process of gentrification creates a sense of displacement among long-term residents. However, some scholars argue that Airbnb and similar platforms enable a model of tourism that is comparably more inclusive, in which small-scale actors participate in the market alongside commercial actors. (Kadi et al., 2019).

The rapid increase in Airbnbs in Salzburg share patterns observed in other European cities. The city placed first in Austria in Airbnb listings per capita during the peak season in August 2019, before the COVID-19 pandemic (Airdna, 2019). The local Airbnb market shows a high degree of commercialization: 76% of hosts operate a single listing but account for only 45% of all listings. The remaining 55% of listings are hosts listing two to four properties (27%) and five or more properties (28%). These actors are referred to as commercial hosts and dominate the STR landscape of Salzburg (Grisdale, 2019).

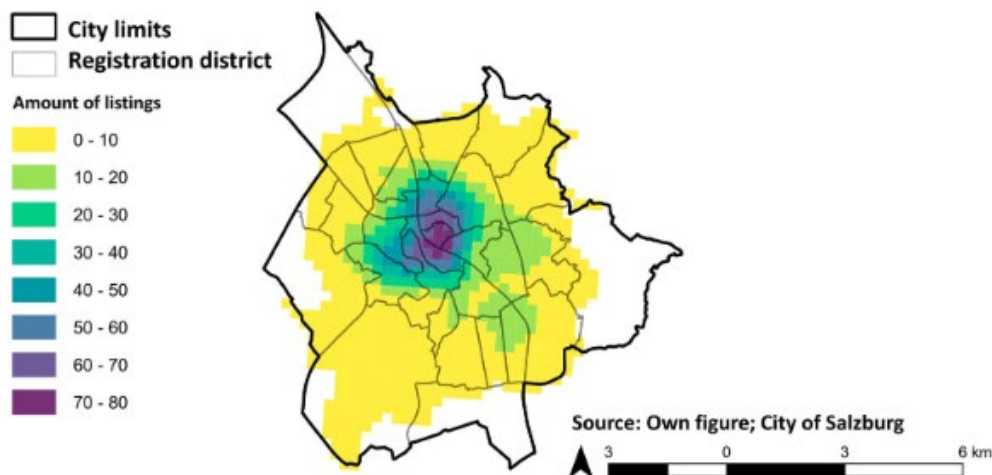


Figure 4: Spatial distribution of Airbnb listings in the City of Salzburg (Smigiel, 2023)

Impacts on local housing market

Multiple studies have shown that Airbnb puts pressure on local housing markets by forcing regular (long-term) housing units away from the market (Grisdale, 2019; Gurran and Phibbs, 2017; Mermet, 2017). Furthermore, there is evidence that Airbnb has a highly uneven distribution and mainly impacts city center areas. The spatial structure of the spread of Airbnbs in Salzburg confirms this finding. Airbnb does not primarily impact all housing markets but exerts high pressure of specific parts of the local housing markets, which in turn is likely to affect broader segments and housing prices in the long term (Barron et al., 2017; Duso et al., 2020). Figure shows the differing share of Airbnb listings to total housing stock. Airbnb is concentrated in the center of Salzburg and some inner neighborhoods compared to the outskirts which suggests that Airbnb is creating significant pressure in specific regions, not the entire city.

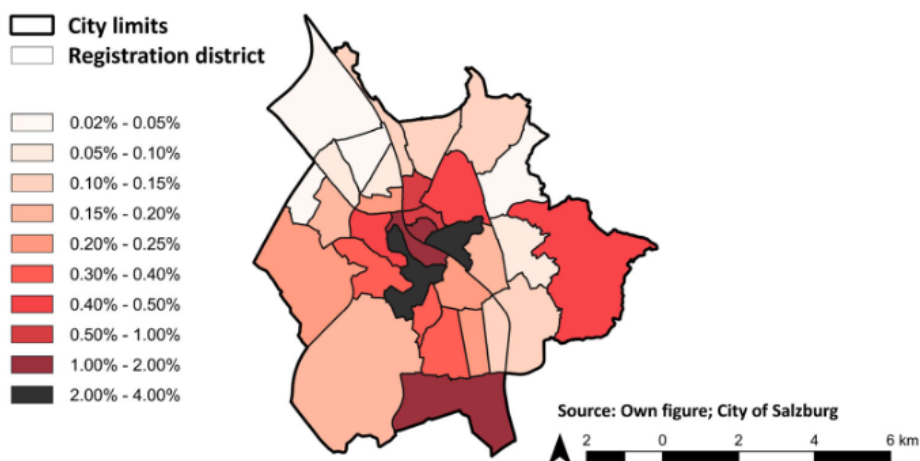


Figure 5: Share of permanently rented entire Airbnb listings as % of total housing stock (Smigiel, 2023)

- **High-Density Areas (dark red/black tones)**
In areas close to the city center, the ratio of Airbnb listings to the housing stock ranges from 1.00% to 4.00%. This suggests that a significant portion of the local housing stock in these areas is devoted to short-term rentals.
- **Medium-Density Areas (red-orange tones)**
The Airbnb influence is also evident in the surrounding neighborhoods, with ratios ranging from 0.20% to 1.00%. These areas are closer to the city center but have less tourist attraction.
- **Low-Density Areas (light beige tones)**
Short term rental share remains below 0.05% in the northern and western parts of the map. These areas may be predominantly residential and have low tourist appeal.

Commercialization and Rent Gap Effects

Due to the commercialized nature of the short-term rental market, Salzburg has a notably high percentage of permanently withdrawn Airbnb listings. Up to 4% of the housing stock in some city center neighborhoods is set aside exclusively for Airbnb rentals; this percentage is similar to that of highly commercialized and touristed cities like Toronto or New York (Grisdale, 2019; Wachsmuth and Weisler, 2018). Kadi et al. (2019) argue that despite thresholds of 60 or 120 days are frequently used to categorize permanent STR activity, they may skew the data to understate the actual extent of market withdrawal. For the long-term housing market, even listings that are available year-round but not booked for 60 days constitute a net loss. 50% of all home listings in Salzburg are available for more than 240 days a year and 77% of all listings are available for at least 120 days. This shows a trend towards the city becoming an all-season tourist destination with a year-round business model. A gap between the potential revenue from STRs and traditional rental income is emerging in Salzburg according to both qualitative and quantitative data. In central areas the trend is even more noticable with STR income reaching up to 300% higher than that of traditional, long-term rentals (roughly €3,000 more per month).

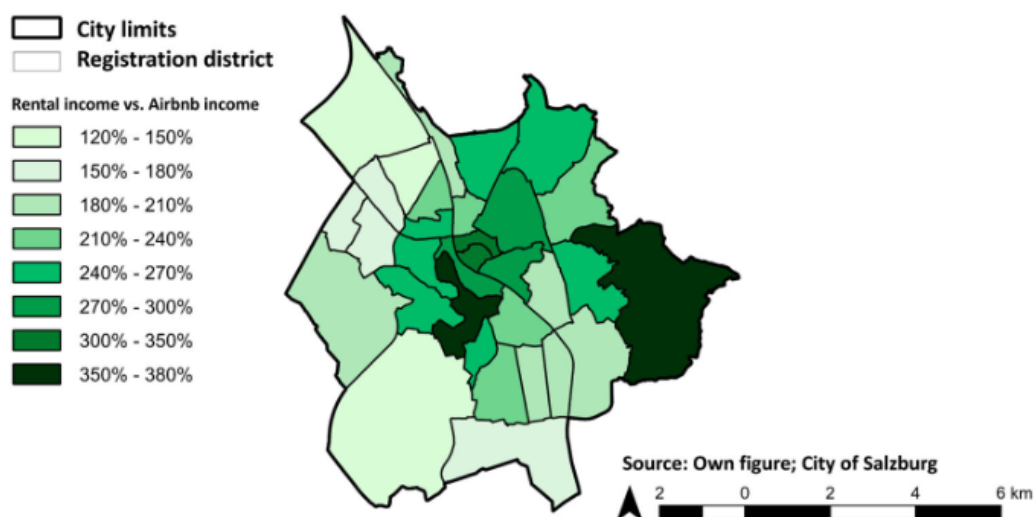


Figure 6: The ratio of monthly long term rental income vs. monthly short term rental income (Smigiel, 2023)

In terms of offering more profit potential, Salzburg's short-term rental market is also incredibly unevenly distributed geographically, as the figure illustrates. Investors and property owners are encouraged to construct short-term housing in the city center due to the high rent difference, which limits the supply of long-term tenants and raises rental prices.

Financialization

Short-term rentals contribute significantly to Salzburg's economic growth, however, they also worsen the existing socioeconomic inequalities by increasing rent prices, disproportionately impacting those already experiencing financial problems. This pattern is consistent with global trends of gentrification, tourism and the financial of real estate. The increasing share of STRs and investor interest drives up property prices.

Housing in Salzburg is transforming from a necessary social service into a financial opportunity that is taken advantage of by entrepreneurs and commercial hosts due to the aforementioned economic factors as well as political factors (Schernthaner, 2024). Land-use restrictions and lack of available land for development constrain new construction, particularly in the city center, while the increasing share of STRs and investor interest drives up property prices. The result is new residential construction occurring in touristic areas like Salzburg Land.

Lack of new construction in central areas leads to Salzburg's suburbs and rural areas experiencing faster population growth than the central areas as most of the employment opportunities are located in central areas. This spatial mismatch caused the average commute of workers to the city center to increase significantly. According to Schernthaner (2024), uneven urban development and increased share of STRs reveal that housing policy is increasingly affected by economic factors and is in line with neoliberal politics rather than addressing socioeconomic inequality.

Covid-19 Pandemic

2020

In 2020, Salzburg's housing market showed stability despite the COVID-19 pandemic. Real estate transactions slowed down in the spring due to uncertainty and lockdowns, but housing prices did not drop. The number of sales decreased by approximately 9% from March to May 2020 due to lockdown. The average free-market apartment in Salzburg province went up to € 5,143 per square meter, a 5.3% increase, while the increase in average detached and semi-detached homes were much higher at 13.1%, reaching € 4,872 per square meter. Salzburg now boasts the second most expensive housing market in Austria after Innsbruck. Average house price is € 545,000 and apartment prices in the city center reach € 5,230/m². Housing increasingly becomes an investment option that is able to resist unexpected phenomena such as the pandemic. In 2020, Austria's national house price index increased by 9.98% (Salow, 2021).

2021

The upward trend in housing prices in Salzburg accelerated in 2021 despite the pandemic. Housing demand was driven by record-low interest rates and people's desire for larger, more private areas that is the result of the pandemic. 12.5% annual increase, which was a record for Austria's national house price index. Demand for detached homes with gardens shot upwards and the focus of buyers was shifted from central apartments to suburbs and rural areas. The shortage of supply in the housing market kept prices rising with double-digit increases seen in Salzburg for both homes and apartments. The number of existing hopes rose 13.1% in Austria in 2021, a year of record levels of expansion for the real estate market of Salzburg (Engel & Völkers).

2022

Housing prices continued to rise in 2022, especially in the first half of the year, despite the lifting of most pandemic related restrictions and the start of normalization. Real estate investment became more and more popular as an investment safe from inflation. Despite a roughly 5% national increase in housing prices, Salzburg's shortage of supply caused larger amounts of increases regionally. The cost of existing homes in the province of Salzburg increased by 11.6% in 2022. But near the end of the year, the European Central Bank started raising interest rates, and Austria tightened mortgage laws, requiring larger down payments and stricter debt to income ratios. Due to these policies, demand started to decline, and housing prices remained stable before peaking in the last quarter.

2023 and Beyond

Salzburg's housing market started to decline for the first time in years in 2023. The increased activity in prices due to COVID has been offset by increasing interest rates and tighter credit access which resulted in reduced purchasing power. After three years of the transformation of the housing market from a distribution system of a social good to an investment, the market began to stabilize. The pandemic had initially accelerated price growth and demand particularly in detached, suburban and rural housing but the macroeconomic adjustments after 2022 signaled a turning point.

The figure by Engel & Völkers (2021) shows the spatial distribution of residential property values in Salzburg. It can be seen that the housing market is segmented and is shaped by regional factors and investment demands. The central and southern districts—including areas such as Nonntal, Aigen, Morzg, and Leopoldskroner are categorized as prime locations (marked in red), with consistently high and rising property prices, reaching up to €9.8 million in 2021 for luxury homes and over €8,600/m² for apartments. These neighborhoods share characteristics of proximity to the historic city center, scenic environment and high status. All making them attractive to wealthy buyers from within and without the country.

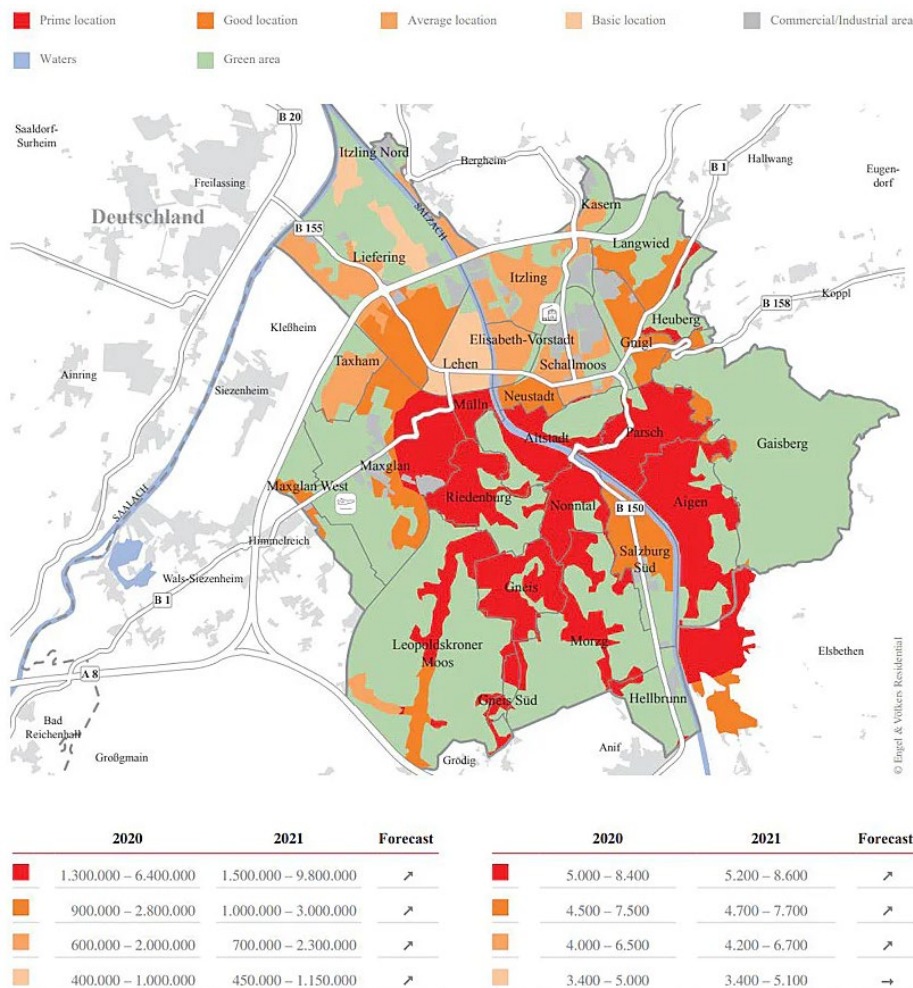


Figure 7: spatial distribution of residential property values (Engel & Völkers, 2021)

In contrast, peripheral and northern areas — such as Liefering, Itzling, and Langwied—are classified as basic or average locations (marked in orange and beige), with comparatively lower property values, ranging from €600,000 to €2.3 million for houses and €3,400 to €5,000/m² for apartments. The prices increased for these neighborhoods in 2020 and 2021 as well, but the rate of growth remained lower than central areas. The forecast arrows across the board show an upwards trend in both house and apartment prices citywide. It can be assumed that Salzburg will remain Austria's second most expensive real estate market after Innsbruck (Engel & Völkers, 2023).

The prime locations shown on figure 7 are concentrated within central Salzburg, surrounded by green areas and limited built-up land shows the spatial inequality in the city. Opportunities for investment in real estate, employment opportunities and increasing prices are all concentrated on high income, inner areas of Salzburg. This segmentation increases the housing prices across the city, increases commute for workers who have been displaced to the peripheries and damages affordability.

In conclusion, Salzburg's housing market has been affected by the pandemic, global economy, regional investment trends and local spatial structure. Rising cost of housing, increasingly unaffordable homes and worsening socio-spatial segregation that is observed today can be traced back to several factors. Housing is becoming an increasingly financialized area, partly

due to the rise of short-term rental platforms such as Airbnb, and the macroeconomic adjustments following the pandemic. This transformation of housing into a commodity driven by land and investor demand along with strict land use regulations affects not only prices but has steadily caused an increase in the commute of workers. Instant increase of popularity in STRs, especially in central Salzburg, has worsened rent differences and decreased the availability of long term housing. The pandemic caused investments despite and partly due to uncertain economic conditions that also reinforced existing trends. Even though the market has been cooled off due to the rising interest rates and stricter lending rules in 2023, structural challenges regarding the housing market remain. Salzburg is not unique in its trends, many cities around the world showed similar patterns with their housing markets becoming more and more defined by market logic rather than the common good. Solving the issues at hand will require a policy shift that focuses on affordability, accessibility, fairness and social well-being in the long term.

Findings from the excursion

During our excursion to Salzburg, we had the opportunity to talk to three experts on the subject of housing in Salzburg. Each of them had a different background, as illustrated below:

First interview: Ing. Mag. Heinz Neumayer

- current head of department 10/03: legal concerns regarding planning, construction, living (= "Referat 10/03 - Rechtsangelegenheiten Planen, Bauen, Wohnen"), which is part of the Salzburg state government.

Second interview: Dr. Andreas Schmidbauer

- department leader of the spatial planning and building authority of the city of Salzburg, (= "Raumplanung und Baubehörde")
- In particular he is responsible for the management of everything that relates to the UNSECO world heritage status of Salzburgs inner city (= "Welterbemanagement")

Third interview: DI Patrick Lüftenegger

- works as a researcher at the Salzburg institute for spatial planning and housing (= "Salzburger Institut für Raumordnung und Wohnen"). Specifically he focuses on housing research.

As our research focused on the driving forces behind housing prices, we prepared three questions regarding different factors that might influence them. During the first interview we had the chance to ask all of them. The latter interviews were structured differently than the first, which resulted in us having less opportunities to ask our questions. Nonetheless, all of them yielded relevant answers for our research questions and are summarized below.

First interview

We met Ing. Mag. Heinz Neumayer in the streets of Salzburg. While strolling through the city, we asked him the three questions we prepared about potential drivers for housing prices in Salzburg City:

1. Vacancy: A classic problem with housing. Hard to measure and regulate. We asked him whether or not he sees potential for a vacancy tax and if it is worth it to try to focus on this problem. Furthermore we wanted to know if he thinks the problem of vacancy is exacerbated in older buildings (Altbauten), due to their rent cap, which lies significantly below market prices.
 - a. Answer: The people who can afford to leave their Salzburg apartments vacant can also afford a vacancy tax. Therefore making it not worth the effort. He also said that he believes vacancy is in fact more of an issue in older buildings (Altbauten).
2. A potential challenge for property developers is the fact that they are often obligated to build new parking spaces for every new residential unit they create, making small apartments, which clearly are in high demand, more expensive and difficult to build in a dense city such as Salzburg. How is Salzburg reacting to this issue?
 - a. Answer: This doesn't pose a problem for Salzburg specifically because the Salzburg state building law allows for municipalities to make changes in the obligations for parking space.

This answer from Ing. Mag. Neumayer proved to be wrong. Salzburgs obligatory parking spaces are defined in attachment 2 of the Salzburg Technical Building Regulations 2015 (= "Anlage 2 der Salzburger Bautechnikverordnung 2015"). These rules can however be changed by the municipalities of Salzburg in accordance with § 38 para. 3 of the Salzburg building code 2015 (= "Salzburger Bautechnikgesetz 2015"), as mentioned by Ing. Mag. Neumayer. However the relevant Salzburg law, called parking space ordinance (= "Salzburg Stellplatzverordnung 2016"), does not make any changes when it comes to obligated parking space for housing (= "Wohnbauten"). To confirm, we sent an E-Mail to the spatial planning and building authority of the city of Salzburg, (= "Raumplanung und Baubehörde"). They replied:

"[...] Sie haben völlig korrekt recherchiert. Für die Stellplatzberechnung [für Wohnbauten] ist die Anlage 2 des BautTG heranzuziehen. [...]"

Translation:

"[...] You have researched correctly. To calculate the obligatory parking space [for housing], attachment 2 of the Salzburg Technical Building Regulations is to be used. [...]"

This is in direct contrast to Ing. Mag. Neumayers Statement. Thus our initial intuition was correct and the laws for obligatory parking spaces make smaller apartments indeed much more difficult and expensive to build in Salzburg City. Specifically, for housing, it is required to build 1.2 car parking spaces (rounded up) and 2 bicycle parking spaces for every new unit built. Therefore when a developer wants to build new apartments with a total of 900 m² of living area in Salzburg, the choices he or she has look something like this:

30 m ² apartments	60 m ² apartments	90 m ² apartments	required car parking spaces	required bicycle parking spaces
30	0	0	36	60
16	7	0	28	46
8	8	2	22	36
5	5	5	18	30
0	6	6	15	24
0	0	10	12	20

As illustrated, building 30 small units requires 36 car parking spaces, while 10 large units only require 12. The extra costs this creates vary from project to project. A generalized statement about the impact of this law is therefore difficult to make. However, given that Salzburg City has various geographical and political factors that heavily limit the available space (i.e. the river Salzach, two hills within the city, strict green space protection etc.) it suffices to say that building new parking spaces, especially for cars, is often limited to underground solutions. These are expensive. In addition with higher m² building costs and potential other factors such as lower demand from owner-occupiers, obligatory parking spaces are likely a significant driver behind the low supply of small apartments in comparison to their demand on the rental market. Further research on the specific impact of obligatory parking spaces on the costs that come with building new small residential units might be suggested.

3. Our third question was about the non touristic short term rental market. We knew that Salzburg takes heavy action against AirBnb and other touristic forms of rental models. However we wanted to ask about whether or not Salzburg is concerned with 1 to 6 month short term rents, which are typically taken by people who need housing due to a temporary, work related change of location. The law regulating this isn't a state but a federal law (MRG § 1 para. 2 Pt. 3b), which is why we wanted to know whether or not Salzburg takes issue with it.
 - a. According to Ing. Mag. Neumayer the state of Salzburg doesn't have a problem with this legislation.

Second interview

The day after our interview with Ing. Mag. Neumayer, we met Dr. Schmidbauer in the Salzburg City Council Building Authority (= "Magistrat Salzburg Baurechtsamt"). The three main takeaways from his short presentation and Q&A afterwards were:

1. protected green space (= "Geschütztes Grünland"):

Dr. Schmidbauer gave us a handout of his presentation, which included the following map:

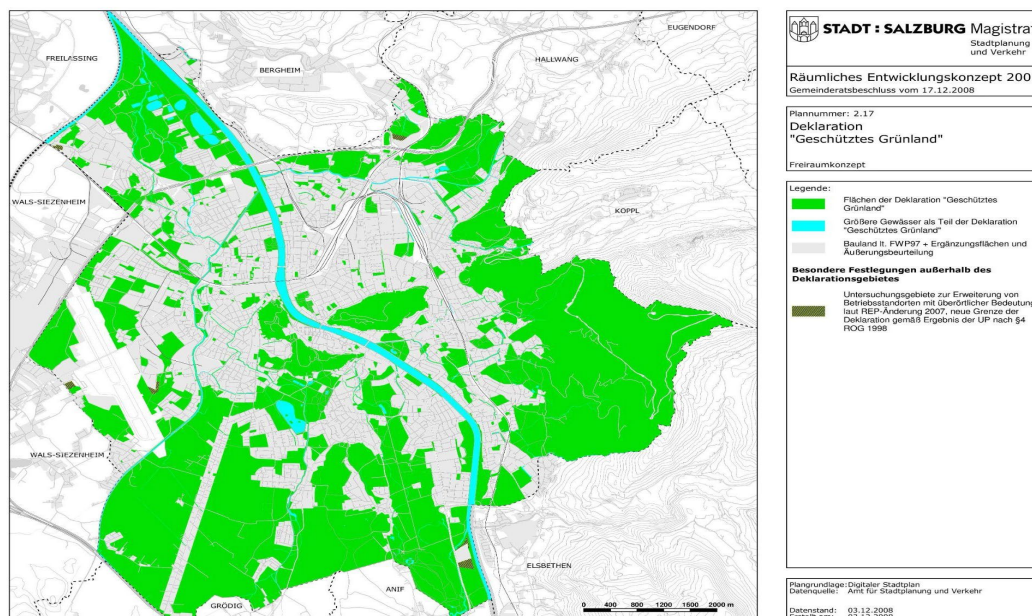


Figure 8: Protected green space (Räumliches Entwicklungskonzept 2007)

It shows all of the land in Salzburg City declared as protected green space (= “Geschütztes Grünland”). He explained that this category of green space has its origins in 1985, when it was first declared by the City Council of Salzburg through the green space declaration (= “Grünlanddeklaration”). Dr. Schmidbauer described the policy as “very restrictive.” He stated that the areas designated as protected green space are “practically impossible to re-designate” and that it represents the “most restrictive green space protection in Austria”.

He further specified that changes to the declaration can only be made when $\frac{3}{4}$ of the City Council vote in favor of it. This only regards relatively small changes however. When the City Council wants to remove areas from the declaration without declaring a replacement area, they have to conduct a referendum. This means that any significant changes to the declaration can only be made with a public referendum *and* $\frac{3}{4}$ of the City Council being in favor of it.

Our research on this topic revealed just that. § 13 para. 2a of the Salzburg City Law 1966 (= “Salzburger Stadtrecht 1966”) states that the presence of at least $\frac{3}{4}$ of the City Council members and a $\frac{3}{4}$ majority of the present members is required in order to make changes regarding the protection of City landscapes relevant to the cityscape. § 53a further states:

“[...] Beschlüsse des Gemeinderates über eine wesentliche Änderung des beschlossenen Schutzes der für das Stadtbild prägenden Stadtlandschaften, insbesondere über die Herausnahme von Flächen aus dem davon erfassten Bereich ohne einen weitestgehend gleichwertigen Flächenersatz, sind jedenfalls einer Bürgerabstimmung zu unterziehen. [...]”

Translation:

“[...] Resolutions of the City Council regarding a substantial amendment to the decreed protection of landscapes that characterize the cityscape, in particular the removal of areas from the zone covered by this protection without a largely equivalent replacement area, must in all cases be submitted to a public referendum. [...]”

As far as our understanding of the legalities of the declaration goes, this confirms the statements made by Dr. Schmidbauer. Other sources also confirm this. The need for a $\frac{3}{4}$ majority and a public referendum when taking green space out of the declaration without largely equivalent replacement areas is described on the website of the City of Salzburg (Greifeneder, 2007). There it is also mentioned that roughly 57 % of the total City area fall under the declaration.

This presents a serious constraint on housing development in the City of Salzburg. ~57 % of the city area is practically erased from future development. However, this percentage refers to the total area of the city and not to the areas that would be potentially suitable for future residential construction. Looking more closely at Figure 8, it appears that almost all undeveloped areas in the city of Salzburg fall under protected green space. A relevant question would therefore be what percentage of the land suitable for development is *not* under the protection of the declaration. Based on the impression from Figure 8, the potential for residential construction in Salzburg is extremely limited.

In combination with Salzburg's growing population, the strong economy, problems with vacancies and Airbnb, obligatory parking spaces, etc., the green space declaration almost guarantees rising scarcity of housing and thus rising rent and sale prices. Further research on the impact of this law, as well as on the remaining reserves of land available for construction, is urgently recommended.

2. UNESCO world heritage

The second main takeaway from the interview regarded the status of Salzburg as a UNESCO world heritage site. In particular, this status is held by the old city center of Salzburg. The main law implementing the regulations set by UNESCO is the Salzburg Old Town Conservation Act of 1980 (= "Salzburger Altstadterhaltungsgesetz 1980"). Finding a map that illustrates the UNESCO zone proved to be oddly difficult but according the website of the City of Salzburg it regards the following area:



Figure 9: UNESCO world heritage zone ([unesco.org](https://www.unesco.org))

Interestingly, the Salzburg Old Town Conservation Act defines its zones differently than UNESCO, as illustrated in attachment 1 of the Act (= "Anhang 1"). This highlights some of the complexity that comes with the world heritage status. The zones defined by UNESCO have to be addressed somehow but aren't reflected in the main piece of legislation that implements regulations to the Old Town of Salzburg. Rather large areas that fall under the buffer zone of UNESCO are not included.



Figure 10: protected area according to the Old Town Conservation Act

Dr. Schmidbauer told us about the administrative and political challenges that come with Salzburg holding this status. However the two most important aspects that he mentioned that regard the housing markets were:

- Limits in building height
 - Areas designated as world heritage are subject to strict rules that limit the maximum height/ cubature of buildings.
- expensive renovations and their subsidies
 - Due to various regulations regarding the world heritage site, renovations prove to be difficult and (very) expensive. However, § 17 of the Salzburg Old Town Conservation Act ensures subsidies for such renovations. Therefore, there is a legal entitlement for subsidies for the owners of buildings within the designated zones.

Being a UNESCO World Heritage site presents additional challenges beyond these two aspects for property owners, developers, and the city of Salzburg, thereby indirectly impacting rental and purchase prices. Detailing these would exceed the scope of this report due to the high complexity and multifaceted nature of the regulations. However, since the effects are mainly confined to the city center, which contains much non-residential space and hardly any affordable housing, these are less severe than, for example, those of the green space declaration.

3. active land policy

Dr. Schmidbauer mentioned Salzburg's severely underdeveloped active land policy as one of the key challenges within the Cities housing sector. In this context, he repeatedly stated, "Salzburg is not Vienna." (= "Salzburg ist nicht Wien."). He explained this by noting that

Salzburg has failed to build up significant historical land reserves within the city. Unlike Vienna, which acquired substantial land at very low costs during the interwar period and never divested it, Salzburg, in contrast, relinquished much of its land holdings.

It is widely acknowledged that Vienna's social housing program significantly contributes to the city's overall affordability. The extensive supply of municipal housing (= "Gemeindewohnungen") serves as a crucial safety net, accommodating individuals who cannot afford the comparatively high market rents. Furthermore, Vienna's market rents remain low compared to cities of similar size and economic strength. While the reasons for this are complex, the high proportion of social housing is frequently cited as a regulating and rent-dampening factor.

The lack of municipal land reserves further constrains the city of Salzburg's room for maneuver, similar to the impact of the green space declaration. This suggests that Salzburg will be heavily reliant on private developers for its future housing construction, and must therefore focus on creating effective incentives. Establishing an active land policy as an alternative appears unrealistic, given the current high prices and scarce supply, which is partly attributable to the aforementioned green space declaration. A comprehensive analysis of the city's scope for action regarding an active land policy thus seems advisable.

Third interview

On the same day as the second interview we met DI Patrick Lüftenegger in the Salzburg Institute for Spatial Planning and Housing (= "Salzburger Institut für Raumordnung und Wohnen").

He gave us a presentation of the 2021 analysis of the affordability of the Salzburg housing market (SIR). While this gave us a good insight into the subject, our primary focus lies on the driving forces behind prices in Salzburg, not on the degree of affordability. While this presentation provided a valuable overview of affordability, its focus was less aligned with our specific research question concerning the primary drivers of housing prices.

After the presentation however, we got the chance for another Q&A. One of the discussion points was, just like in the previous interview, the city of Vienna and its housing policy. DI Lüftenegger agreed with us that Vienna is a kind of historical unicorn and that its system of communal housing is not replicable, at least not in Salzburg. Vienna acquired the necessary land for this system at a time of very low prices. Furthermore, the city has been continuously governed by one party for 80 years. Alternating ruling parties, on the other hand, would come with significantly more challenges for such a communal housing program.

However, DI Lüftenegger noted that he sees great potential in non-profit housing developers (= "Genossenschaftswohnungen"). Supported by a Housing Fund, like in Vienna, this would be an adequate and contemporary system for Social Housing.

The question of whether a system based on non-profit housing developers can address the challenges of social housing in Salzburg is complex and not the focus of this report. However, DI Lüftenegger substantiated the challenges in this area mentioned by Dr. Schmidbauer.

This concludes our main takeaways from the excursion. Overall, the first two interviews in particular gave or led us to some key price driving factors. Our external research part ends with this and this marks the beginning of a deeper dive into the dataset we've been given.

The basics of the data

Key figures

At the beginning of our data analysis, we would like to present some basic key figures that can be derived from the provided and cleaned data set. These fundamental key figures about the housing market in and around Salzburg already provide some very important information and reflect many of the factors emerging from our research.

The two tables below show some average values from the data set. For our research purposes, the key figures of the absolute price, price relative to size, and the listing time were particularly interesting. Prices are obviously of direct research interest for this report, and listing time reflects the level of demand quite well.

Apartments (surrounding region)	
Average sale price	€ 462.336
Average rent price	€ 1.264
Average size	79 m ²
Average sale price per m ²	€ 5.364
Average rent price per m ²	€ 15,3
Average listing time for sales	48 days
Average listing time for rentals	21 days
Highest sale price	€ 1.520.000
Highest rent price per month	€ 4.108

Tab. 2: Apartments Salzburg region (own calculation)

Apartments (Salzburg City)	
Average sale price	€ 470.749
Average rent price	€ 1.458
Average size	76 m ²
Average sale price per m ²	€ 6.061
Average rent price per m ²	€ 18,4
Average listing time for sales	49 days
Average listing time for rentals	24 days
Highest sale price	€ 1.490.000
Highest rent price per month	€ 4.180

Tab. 1: Apartments Salzburg City (own calculation)

To begin with, it should be noted that the purchase prices, in particular, only represent asking prices and not actually paid prices. Apartments for sale are almost always listed online at a higher price than the actual achieved sales price. Rental prices, on the other hand, offer much less room for negotiation. The figures must therefore be viewed in this context.

Since this data is still very basic and hardly allows for any conclusions about price drivers, other than that a location in Salzburg City generally leads to a higher price, only the most important differences between these will be briefly highlighted. To provide a better overview, we have shown the percentage deviation of the above values in the following table. This always refers to how much higher or lower the respective figure was in Salzburg City than in the surrounding regions.

Only four of the values we researched deviated by more than 5% between Salzburg City and the surroundings. These were, however, significantly higher and each in the double digits. Without wanting to draw premature conclusions, it can be stated that the three deviations in the price-related key figures were to be expected. Apartments in Salzburg City had on average almost 20% higher square meter rent and 13% higher sales prices than those in the surroundings. Absolute rental prices were also higher.

Salzburg City compared to surroundings	
Average sale price	1,8%
Average rent price	15,4%
Average size	-4,4%
Average sale price per m ²	13,0%
Average rent price per m ²	19,7%
Average listing time for sales	1,5%
Average listing time for rentals	14,0%
Highest sale price	-2,0%
Highest rent price per month	1,8%

Tab. 3: Comparison City to surroundings (own calculation)

Interestingly, the absolute purchase prices, on the other hand, were not even 2% higher. This is undoubtedly also related to the average 4.4% smaller apartment size in Salzburg City, but the discrepancy is worth mentioning. The last, significantly higher key figure was listing time for rental apartments. Rental apartments in the surrounding regions thus apparently showed an even higher demand than in Salzburg City itself. This could be related to the lower prices. People who cannot afford Salzburg City but work or study there are forced to move to the surroundings.

Size categories

One way to explore the price dynamics of the housing market in Salzburg is by categorizing apartments in size. Apartment size is usually the biggest influencing factor on its price, given a similar location. Different apartment sizes also reflect demand from different groups of people. This is why we chose to analyse the average prices for different apartment sizes, as seen in the figure below.

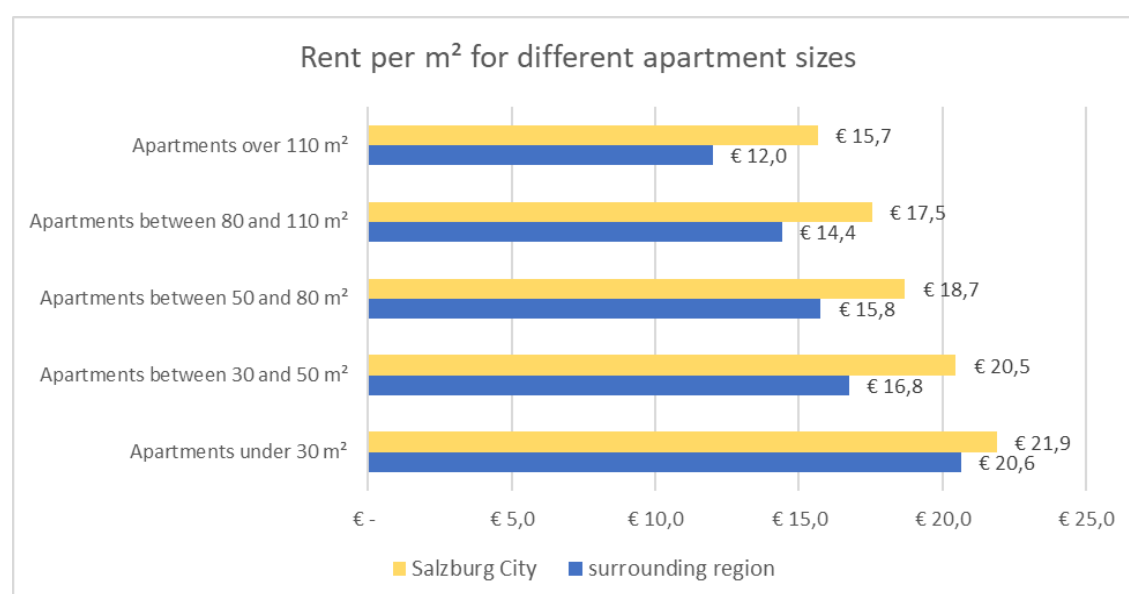


Figure 11: Rent per m² for different apartment sizes (own calculation)

Doing this revealed a very clear trend. The smaller the apartment, the higher the rent per m². The difference in average rent between the biggest and the smallest apartment category was

about 39 % in Salzburg City and a striking ~72 % in the surrounding region. This discrepancy becomes even more striking when considering that a large portion of apartments over 110 m² are likely luxury apartments, whereas those under 30 m² are predominantly entry-level or student accommodations, frequently older and with (much) more basic amenities. This stark contrast in quality and target demographic further amplifies the observed price per square meter inverse correlation.

Another interesting note is that this trend seemed to be more pronounced in the surrounding regions of Salzburg, especially for the smallest apartments. Once again, the reason for that might be affordability. The gap between Salzburg City and the surrounding areas clearly was lowest within the smallest apartment category. All other apartment categories defined by us had a visibly much larger gap. This could reflect an extreme demand for small, 1 bedroom apartments within both Salzburg City and the surrounding regions, resulting in their rents being very similar in both locations.

The next logical thing would be to look at sale prices. Following the clear trend in rents, sale prices should follow at least roughly the same trend. As demonstrated in the chart below, they did not.

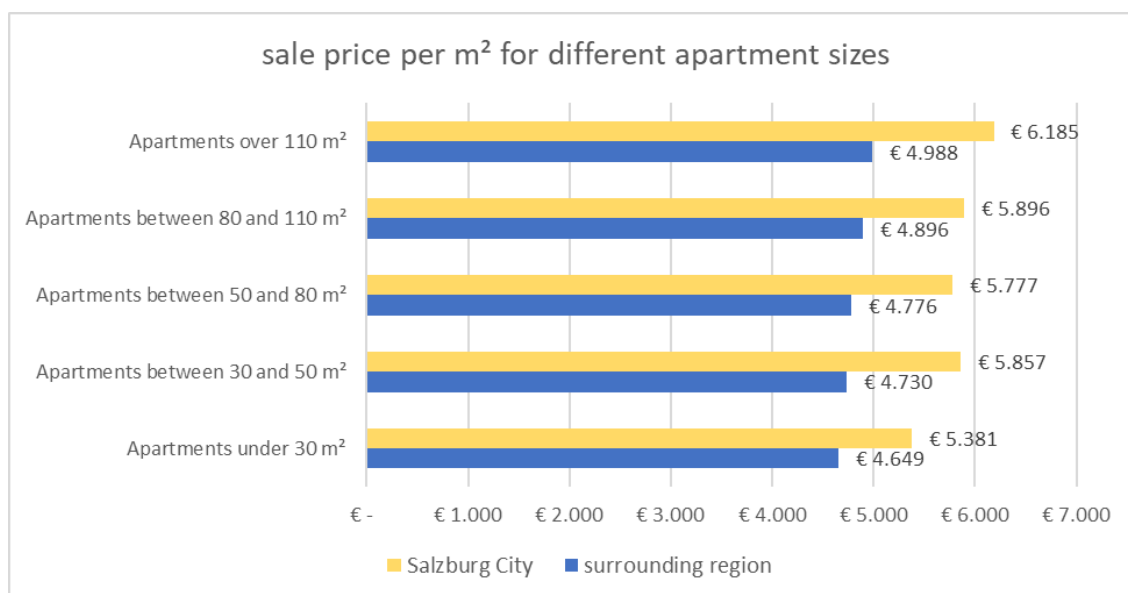


Figure 12: Sale price per m² for different apartment sizes (own calculation)

The trend for sale prices is actually almost exactly opposite. For both Salzburg City and the surrounding regions, the smallest category of apartments actually had the lowest average sale price, while the largest category had the highest average sale price. For the surroundings, all other categories increased in m² price as size increased. Almost the same is the case for Salzburg City, except that the middle category (50 - 80 m²) breaks the trend. That being said, the middle three categories in Salzburg City had virtually the same average m² price as they varied by less than 1 %.

This represents a staggering discrepancy. This counterintuitive finding can be attributed to several factors identified in our research:

- **Regulatory Constraints on Supply:** Salzburg's Technical Building Regulations requiring 1.2 car parking spaces per unit make the construction of new, small apartments disproportionately expensive and thus less common. This leads to an older, less desirable building stock in the small-apartment category, which in turn suppresses the average sales price, despite the intense demand on the rental market.
- **Divergent Demand Profiles:** A very high demand from renters but a low demand from owner-occupiers. Owner-occupiers need substantial equity to buy an apartment and therefore tend to be wealthier and are more likely to look for larger apartments. This could lead to a diminished demand for small apartments from owner-occupiers.
- **Influence of the Luxury Segment:** Larger apartment categories contain a more significant luxury segment, which inflates their average square meter sales price and contributes to the opposite trend.

Another noteworthy aspect with sale and rent prices is that the values between Salzburg City and the surrounding region were once again closest to each other in the smallest category.

It is very clear that the size of an apartment inversely drives up rent prices per m². As far as sale prices are concerned however, this trend seemed to be completely reversed. Given this outstanding discrepancy, further research is definitely recommended.

Looking at average listing times for the same apartment size categories paints a very similar picture as prices.

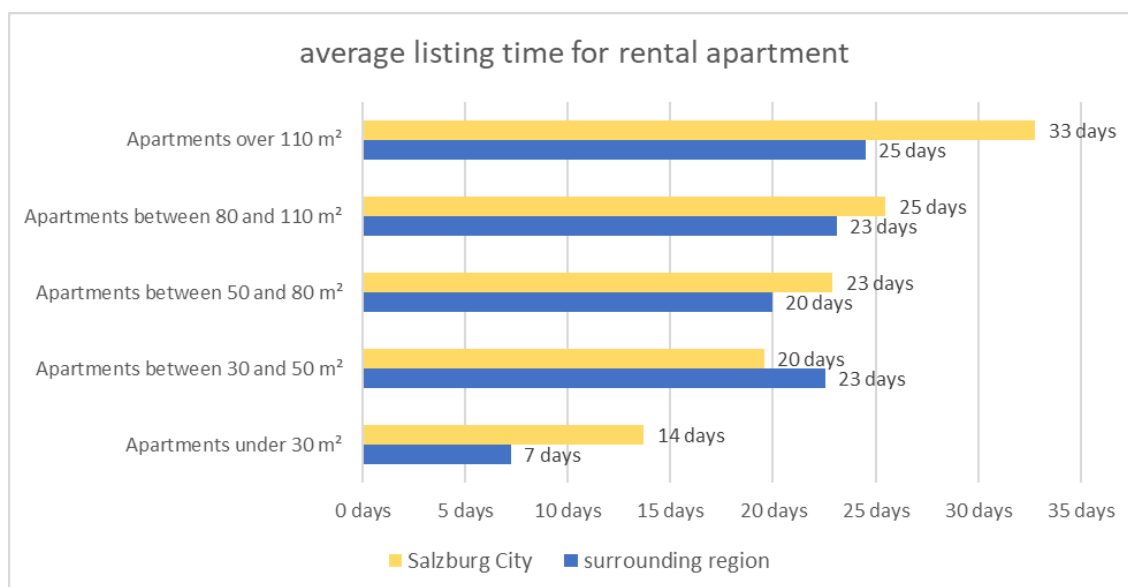


Figure 13: Average listing time for rental apartment (own calculation)

Rental apartments in Salzburg City and its surroundings tended to stay on the market for a shorter period, the smaller they were. While the same trend existed for rental prices, the one found for listing time was far from as linear. The three middle categories in both Salzburg and its surroundings differ only slightly in average listing time. The values for the largest and, especially, for the smallest categories are extreme. Apartments under 30 m² remained on the

market for an average of only two weeks in Salzburg City and even just one week in the surroundings. The latter represents a nearly 70% shorter market duration than in the next higher category. The largest apartments, on the other hand, stayed on the market significantly longer than the second largest, particularly in Salzburg City, specifically by 32%.

While the continuing trend of strong demand for very small apartments was to be expected, this data nonetheless paints an extreme picture. The average listing times for sale apartments on the other hand, again, showed a different market behavior.

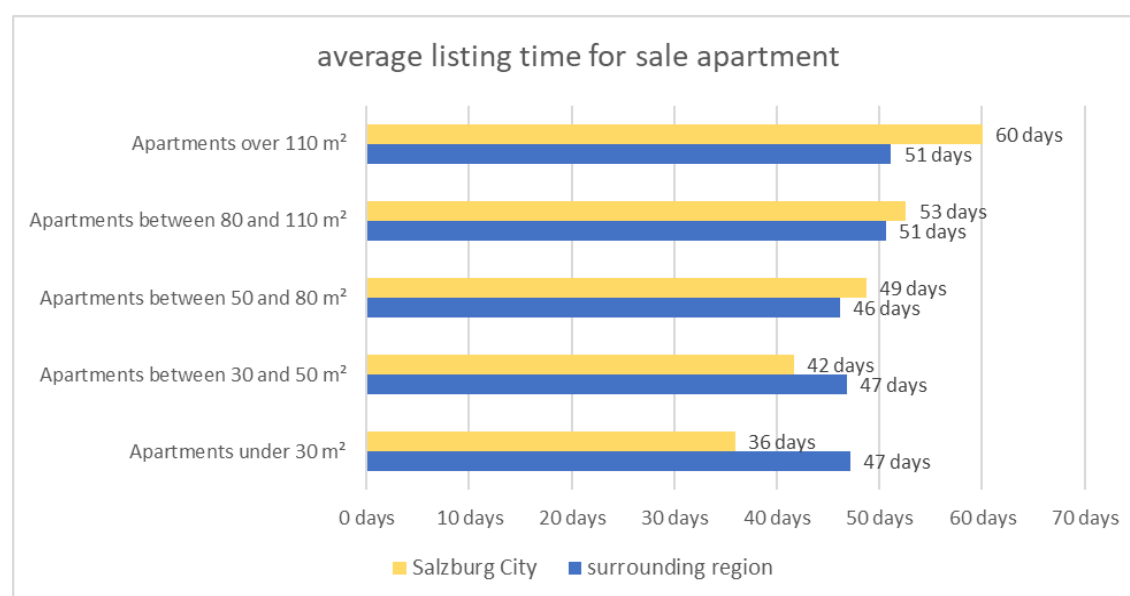


Figure 14: Average listing time for sale apartment (own calculation)

Listing time for sales apartments in general was much higher than for rental apartments, which shouldn't come as a surprise. The averages from Salzburg City also reflected a very similar trend than the average listing time and price for rental apartments did. The largest and smallest categories were not as extreme though. An average of 36 days in listing time for apartments under 30 m² represents only a ~14 % shorter average than the next category in line. For rentals, this difference was 30 %.

Conversely, the average listing time for sales apartments in the surroundings don't reflect a clear trend at all. The averages were very similar and the lowest value was actually found in the middle category. The largest and smallest apartments had tied values with the next category in line. This is somewhat similar to the average m² prices for sales apartments in the surrounding region. They depicted a trend but all values were also rather similar to one another, with the highest and lowest value being only 7.3 % apart from each other. For listing time, the lowest average of 46 days and the highest of 51 days are 10.9 % apart, for a range of 5 days. This is significant to note, as the range was 24 days for apartments in Salzburg City, close to five times as much.

Conclusions

Price Discrepancies by Location: Apartments in Salzburg City generally command higher prices, with approximately 20% higher square meter rent and 13% higher sales prices on average compared to the surrounding regions.

Rental Market Dynamics: There is a clear inverse relationship between apartment size and rent per square meter; smaller apartments have significantly higher per-square-meter rents. This trend is more pronounced in the surrounding region, where the smallest apartments show an extreme demand, leading to minimal price gaps between Salzburg City and its surroundings for this category. Listing time for rental apartments also generally decreases with smaller sizes. Demand for rental apartments appears generally higher in the surrounding regions.

Sales Market Peculiarities: Contrary to the rental market, sale prices per square meter show an almost opposite trend; the smallest apartment categories tend to have the *lowest* average sale price per square meter, while the largest have the highest. This is possibly influenced by factors such as the age of the housing stock, demand from owner-occupiers, and the presence of a luxury segment in larger apartments. This outstanding discrepancy warrants further research.

SPATIAL REGRESSION ANALYSIS OF RENTAL AND SALES APARTMENT PRICES IN THE SALZBURG REGION

Introduction

This study explores spatial dynamics in housing markets by analyzing rental and sales apartment prices across the Salzburg region. Using spatial econometric models, we identify key socio-demographic and structural determinants while comparing the spatial dependency patterns in both markets.

Method

Spatial regression analysis for discovering spatial dependence

Regression analysis is one of the most effective tools to **quantify the relationship** between housing prices and their influencing factors. It allows researchers to identify key drivers (Rosen, 1974), estimate marginal effects and control of confounding variables. Traditional OLS (Ordinary Least Square Estimation) assumes independent and identical distribution among all observations. The basic form of an OLS model is:

$$Y = X\beta + \varepsilon$$

where Y is the dependent variable, X is the matrix of explanatory variables, β is the vector of parameters, and ε is the error term.

However, in spatial data analysis, the assumption is frequently violated due to the existence of spatial autocorrelation, which means spatial statistics always influence each other and tend to be more similar than the distance one (McMillen, 2003). Tobler's First Law of Geography also mentioned that Everything is related to everything else, but near things are more related than distant things (Tobler, 1970). If the spatial dependence is ignored, OLS estimate may eventually lead to a biased or inefficient estimation of model index and overestimate the effectiveness of model (Anselin, 1988).

To address spatial dependence, **spatial autoregressive models** are typically employed, particularly the **Spatial Lag Model (SLM)** and the **Spatial Error Model (SEM)**. The Spatial Lag Model extends the standard regression mode by including a spatially lagged dependent variable, and the Spatial Error Model handles Spatial Autocorrelation by defining a spatial autoregressive error process (Anselin, 1988).

The Spatial Lag Model (SLM) introduces a spatially lagged dependent variable:

$$Y = \rho WY + X\beta + \varepsilon$$

Where W is the spatial weights matrix, WY is the spatial lag of the dependent variable, and ρ measures the strength of spatial dependence. The SLM assumes that the value of the dependent variable in one location is influenced by the values in neighboring locations.

In contrast, the Spatial Error Model (SEM) assumes that the spatial dependence is in the error term:

$$Y = X\beta + u, u = \lambda Wu + \xi$$

Here, u is the spatially autocorrelated error, λ is the spatial autoregressive coefficient in the error term, and ξ is the i.i.d. disturbance. SEM is appropriate when the spatial correlation arises from omitted spatially structured variables or spatial measurement error (LeSage & Pace, 2009).

Ordinary Least Squares (OLS) regression was first conducted, to determine which model is more suitable, **Lagrange Multiplier (LM) tests** are used to detect spatial dependence. Both the LM-lag and Robust LM-lag statistics were significant ($p < 0.001$), whereas the LM-error and Robust LM-error were not. This suggests that spatial lag dependence is dominant. Therefore, a **Spatial Lag Model (SLM)** was adopted for further analysis.

Spatial autocorrelation analysis for discovering spatial heterogeneity

Spatial autocorrelation refers to the correlation of the same variable at different spatial locations, serving as a method to measure spatial dependence. **Local indicators of spatial association (LISA)** quantify the degree of correlation between each spatial unit and its neighboring units for a given attribute (Anselin, 1995). **Global spatial analysis** (such as spatial regression analysis) assumes spatial stationarity and considers that the entire area follows one of three distributional trends: clustering, dispersion, or randomness. However, it does not indicate which specific units contribute most to the global spatial pattern. Within a study area, different local regions often exhibit varying degrees and types of spatial autocorrelation, a phenomenon known as **spatial heterogeneity** (Fotheringham et al., 2000). By constructing local spatial autocorrelation indicators, it becomes possible to accurately capture the heterogeneous characteristics of spatial elements across the entire area.

In this study, we focus on whether the rental and for-sale apartment markets around Salzburg exhibit signs of **polarization**, namely, whether high-price and low-price areas show spatial clustering. Due to the presence of isolated units in the study area, a **k-nearest neighbors (KNN)** spatial weight matrix was adopted instead of the **Queen contiguity (order = 1)** matrix used in the global regression. After comparative testing of different K values, $K = 7$ provided the most robust clustering results and was therefore used throughout the LISA diagnostics.

Cluster Type	Interpretation
High-High	Expensive housing zones clustered → High-end residential areas
Low-Low	Low-price housing zones clustered → Affordable or peripheral areas
High-Low	Price outlier zone in lower-price region → Luxury or gentrifying enclave
Low-High	Underpriced zone in expensive surroundings → Value gap or potential development zone
Undefined	No rental or sales apartments exist in this Katastralgemeinde

Data process

Varieties

To explore the determinants of housing market outcomes, we employed a set of explanatory variables grouped into three categories: **housing characteristics**, **social variables**, and **demographic variables**. The housing characteristics include **average price per square meter** (AvM2Pri), **average unit size** (AvM2), and **average listing duration** (AvDday), all sourced from the House4All database. Social variables capture spatial and socioeconomic dimensions, comprising **distance to the nearest city center** (NEARr_DIST), **employment density** (Emp_Den), **the proportion of non-EU migrants** (NEU_Rat), and **public transport coverage** (Transport), with spatial data derived from OpenStreetMap and Eurostat's 2021 Census grid. Demographic variables include **population density** (Pop_Densit), **the proportion of elderly residents** (G65_Rat), and **net migration rate** (F_in_out_), all obtained from Eurostat's 2021 Census. These variables allow for a comprehensive assessment of spatial, social, and market factors affecting housing prices.

In this study, the distance from a given area to the nearest *Stadtgemeinde* (*Stadt Salzburg*, *Oberndorf bei Salzburg*, *Seekirchen am Wallersee*, *Neumarkt am Wallersee*, *Hallein*) is used as a proxy for the accessibility of public services. A negative relationship is assumed: as the distance increases, access to services such as education, healthcare, and cultural facilities tends to decrease. However, this approach has inherent limitations. While *Stadtgemeinde* status provides a proxy for service access, actual functional centrality may vary. Therefore, while the use of distance to *Stadtgemeinden* provides a useful first approximation, further analysis incorporating actual facility data or centrality indices would enhance the robustness of the results.

Research Unit

In this study, the *Katastralgemeinde* was selected as the basic spatial unit of analysis. A *Katastralgemeinde* represents the smallest legally defined land parcel in Austria, documented in the national cadastre and typically used for property registration and taxation. Unlike larger administrative units such as *Gemeinden* (municipalities), *Katastralgemeinde* units offer much

higher spatial resolution, enabling a more precise representation of urban structure, land use, and neighborhood-level dynamics. This finer granularity allows for a more detailed spatial analysis of housing market behavior, particularly in capturing micro-scale variations in price, demographics, and accessibility that would be averaged out or obscured at the municipal level. Therefore, using *Katastralgemeinde* data supports a more accurate spatial econometric modeling and better reflects localized housing market dynamics.

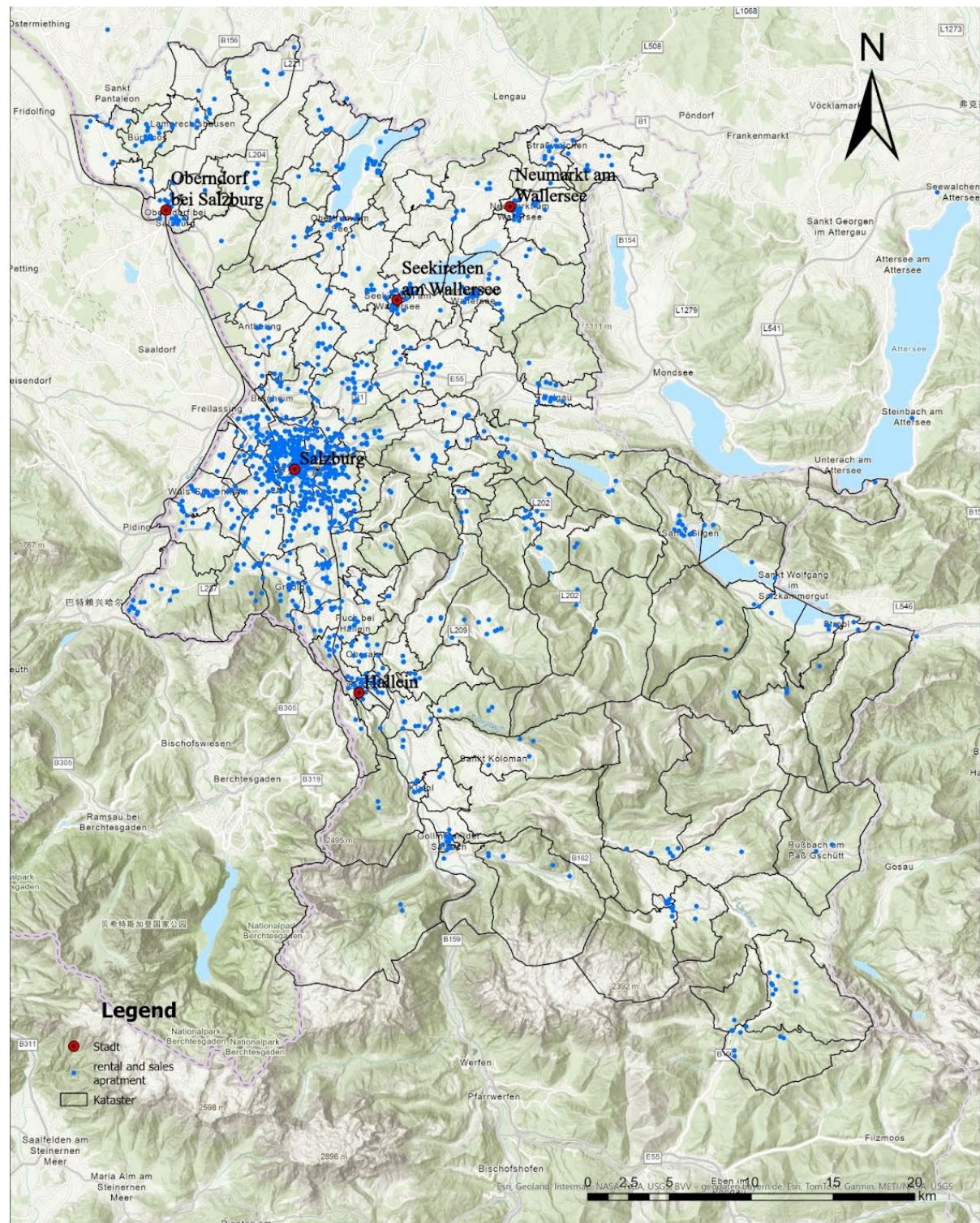


Figure 15: Listed apartments in research area of 134 *Katastralgemeinden* in total, (own representation)

Data pre-process

Housing data, which included georeferenced point-level coordinates of individual listings, was spatially joined to cadastral units. Average price per square meter, average listing days and average size were then calculated for each Kataster for both rental and sale markets.

To enable spatially detailed socio-spatial analysis, population data in raster format (1,000 m resolution) was matched to cadastral units (Katastralgemeinden). Using ArcGIS, the raster grid was spatially overlaid with cadastral boundaries. When a single raster cell intersected multiple Katastralgemeinden, a uniform population distribution within the cell was assumed, and the cell's population was allocated proportionally by area to each intersecting Kataster. While this introduces some estimation error, it provides substantially finer spatial detail than population statistics aggregated at the municipality (Gemeinde) level.

Public transport accessibility was measured as the proportion of the population within 500 meters of transit stops in each Kataster. This was calculated by identifying population raster cells falling within the 500 m buffer zones around public transport stations. For cells that only partially intersected the buffer, population was again proportionally allocated based on the overlapping area.

All datasets were collected from recent years to ensure temporal consistency across variables.

In the current approach, average housing prices are calculated for each Kataster without weighting for data density. As a next step, spatial smoothing or minimum observation thresholds could be applied to improve the stability of estimates, particularly in areas with sparse housing listings. This would enhance the reliability of spatial pattern interpretation.

Regression result

Rental Apartment

The Spatial Lag Model (SLM) for **rental apartment prices (AvPriM2_RA)** reveals a **moderate but statistically significant spatial dependence** ($\rho = 0.300$, $p = 0.0002$), indicating that **rents in one area are positively influenced by those in neighboring areas**. The model explains **41.5% of the variance** in rental prices ($R^2 = 0.415$), and diagnostics show **no heteroskedasticity** (Breusch–Pagan test $p = 0.403$) and **strong support for including the spatial lag term** (Likelihood Ratio $p = 0.00059$).

Several explanatory variables are significant:

- **Positive effects:**
 - Elderly population ratio (G65_Rat, $p < 0.001$)
 - Employment density (Emp_Den, $p = 0.015$)
 - Non-EU immigrant ratio (NEU_rat, $p = 0.100$, marginal)
- **Negative effects:**
 - Population density (Pop_Densit, $p = 0.012$)
 - Distance to nearest Stadtgemeinde (NEAR_DIST, $p = 0.025$)

These findings suggest that **higher rental prices are associated with older populations, job-rich areas, and immigrant communities**, while areas that are more densely populated or farther from regional service centers (Stadtgemeinden) tend to have **lower rents**. Areas with higher proportions of elderly residents tend to command higher rents, possibly reflecting established neighborhoods or service proximity. For detailed regression data please see Appendix A.

Sales Apartment

The Spatial Lag Model (SLM) for **sales apartment prices (AvPriM2_SA)** shows a **strong and highly significant spatial dependence** ($\rho = 0.522$, $p < 0.001$), confirming that **property prices are heavily influenced by surrounding areas**. The model has a **higher explanatory power** than the rental model ($R^2 = 0.503$), though it presents **heteroskedasticity** issues (Breusch–Pagan $p = 0.018$), and the spatial lag term is strongly justified (LR $p < 0.001$).

Key explanatory variables include:

- **Negative effects:**
 - Average listing duration (AvDday_SA, $p < 0.001$)
 - Average unit size (AvM2_SA, $p = 0.051$, marginal)
 - In- and out-migration balance (F_in_out__, $p = 0.018$)
- **Positive effects:**
 - Transport accessibility ($p = 0.083$, marginal)
 - Elderly ratio (G65_Rat, $p = 0.103$, not significant)

These results suggest that **faster-selling units and smaller apartments achieve higher price per m²**, and **well-connected areas attract a price premium**. Migration patterns and market behavior play a key role, reflecting **investment-driven dynamics** in the ownership market. For detailed regression data please see Appendix B.

Spatial heterogeneity analysis

Rental Apartment

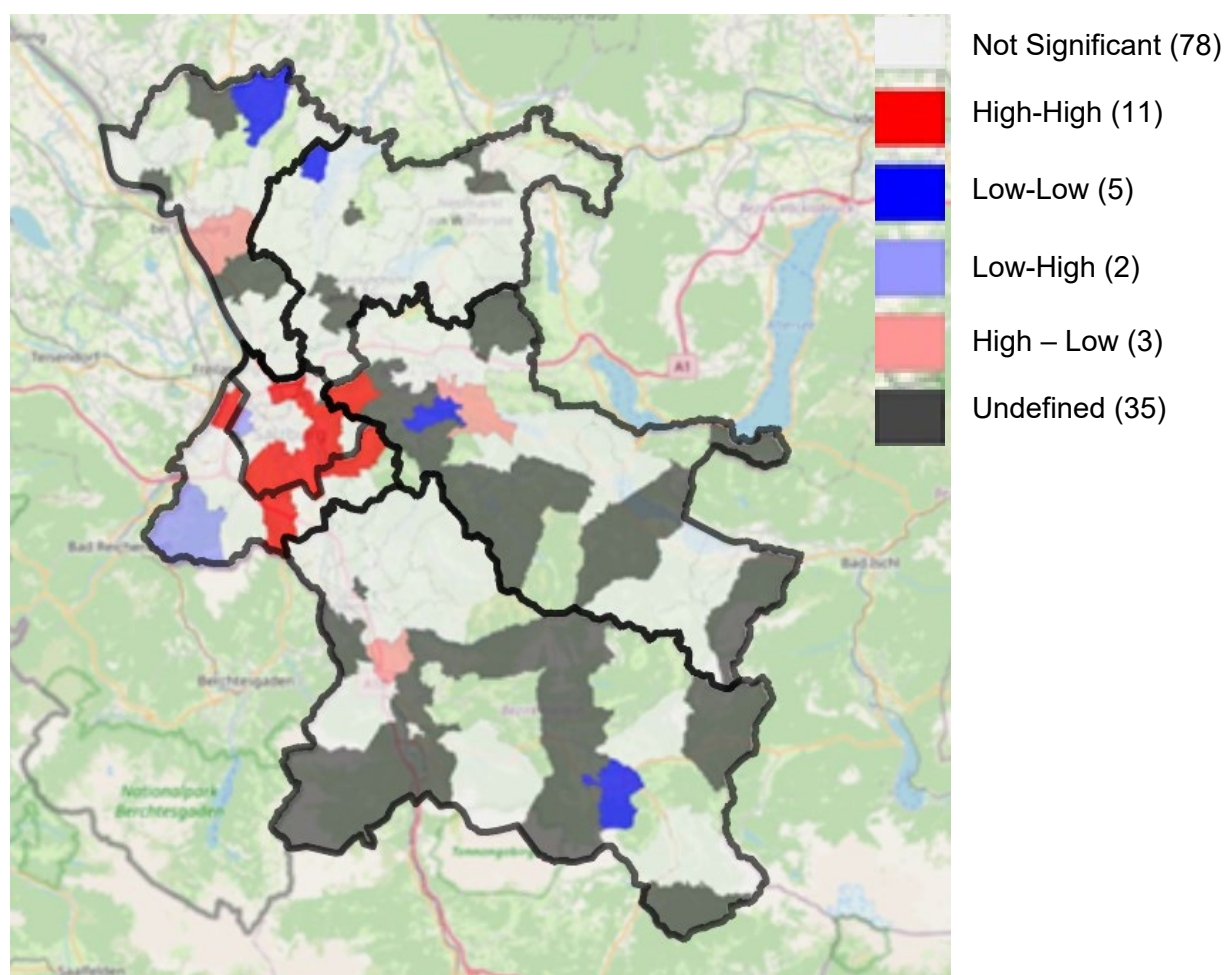


Figure 16: Map for rental apartments created in LISA (own representation and calculation)

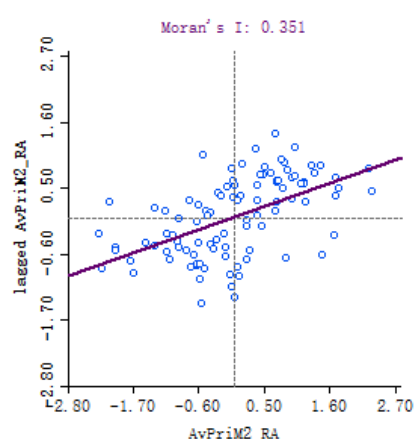


Figure 17: Moran scatter plot for rental apartments ($knn=7$, $p=0.001$), (own calculation)

To explore the local spatial patterns of the rental apartment market in the Salzburg region, a Local Indicators of Spatial Association (LISA) analysis was conducted using average rental prices per square meter (AvPriM2_RA) as the key variable.

The global Moran's I value was **0.351** with a highly significant **p-value of 0.001**, indicating a strong positive spatial autocorrelation in rental prices—areas with similar rental levels tend to cluster geographically. The LISA cluster map further revealed distinct spatial patterns. A total of **11 High-High clusters**—where high-priced units are surrounded by other high-priced units—were predominantly located in and around the city of Salzburg, forming a clear rental price hotspot. In contrast, **5 Low-Low clusters** appeared in peripheral rural or mountainous areas, identifying spatially concentrated low-rent zones.

Additionally, **3 High-Low** and **2 Low-High clusters** indicated pockets of spatial heterogeneity, where units exhibit values that contrast with their neighboring areas—possibly reflecting areas undergoing gentrification or structural transition. The majority of the study area (**78 units**) showed no statistically significant local autocorrelation, suggesting either mixed market conditions or transitional zones. Finally, **35 units** were labeled as undefined due to missing data or insufficient neighbors.

These findings confirm the **existence of spatial polarization** in the rental apartment market, with high-value areas clustering near the urban core and low-value areas distributed along the periphery. The spatial heterogeneity revealed by LISA offers important insights for housing policy, affordability assessment, and urban development planning.

The spatial lag model confirmed the presence of significant spatial dependence in rental apartment prices, with a spatial lag coefficient (ρ) of 0.300 ($p = 0.0002$). This finding is consistent with the results of the LISA analysis, where a Moran's I value of 0.351 ($p = 0.001$) indicated strong positive global spatial autocorrelation. The LISA cluster map further revealed distinct patterns of high-value (High-High) and low-value (Low-Low) clusters, particularly concentrated around Salzburg's urban core and peripheral rural areas, respectively. These clusters visually confirm the spatial spillover effects captured by the spatial lag model.

Moreover, the presence of High-Low and Low-High outliers in the LISA results points to localized spatial heterogeneity that the global regression model does not fully capture, suggesting that further exploration using geographically weighted regression (GWR) or multilevel models may be warranted. Overall, the LISA results substantiate the inclusion of a spatial lag term in the regression model and provide additional insights into the spatial structure of the rental housing market.

Sales Apartment

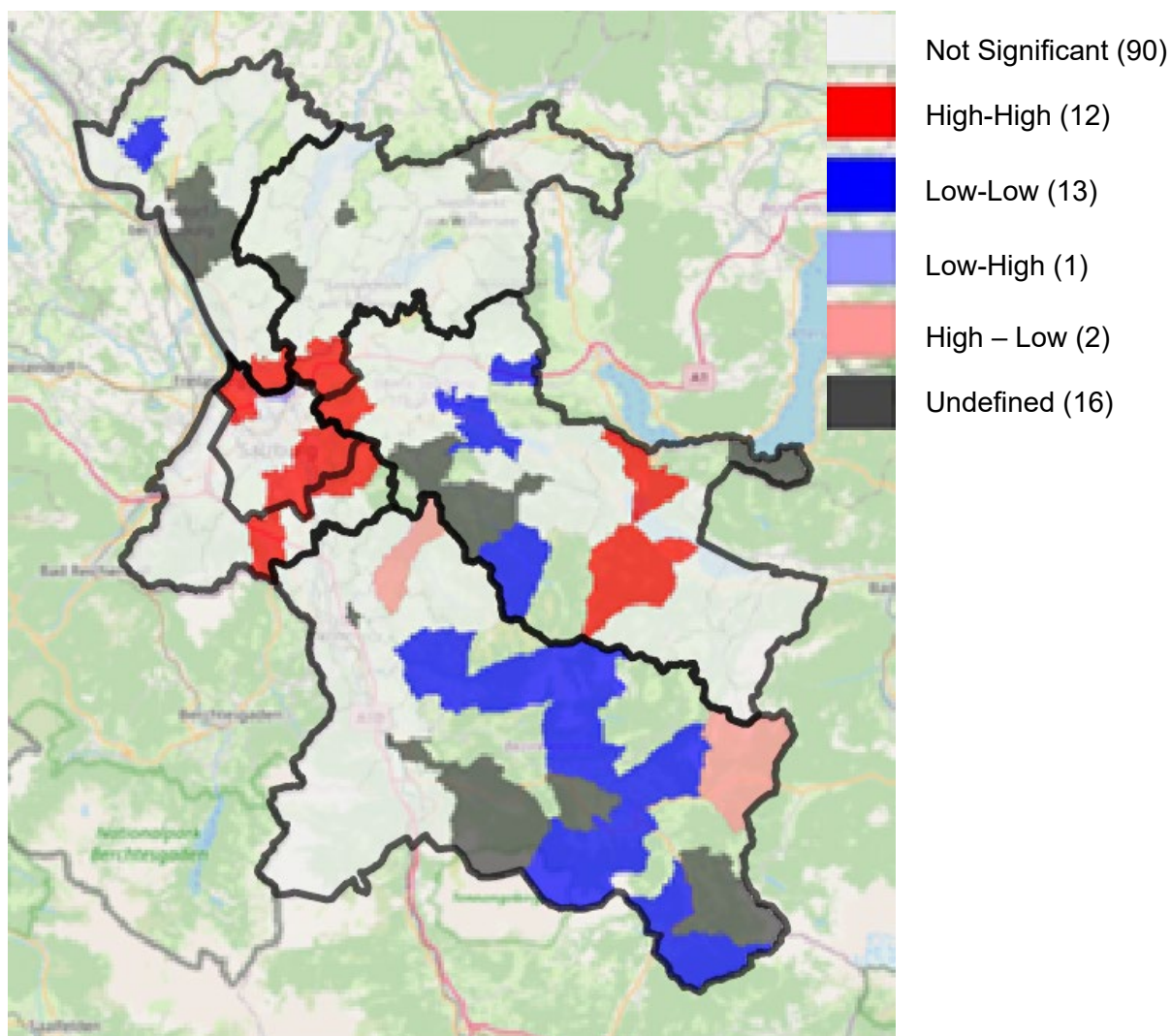


Figure 18: Map for sales apartments created in LISA (own representation and calculation)

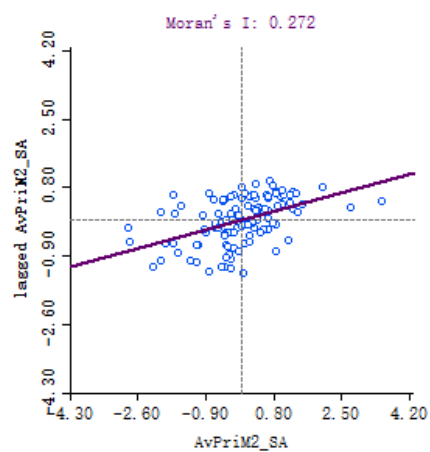


Figure 19: Moran scatter plot for sale apartments ($knn=7$, $p=0.001$), (own calculation)

To examine the spatial clustering characteristics of the sale apartment market in the Salzburg region, a Local Indicators of Spatial Association (LISA) analysis was applied using average sale price per square meter (AvPrim2_SA) as the target variable.

The global Moran's I value was **0.272**, with a statistically significant result ($p = 0.001$), indicating a **moderate but significant positive spatial autocorrelation** in apartment sale prices. This suggests that areas with similar price levels are spatially clustered rather than randomly distributed.

The LISA cluster map reveals more nuanced spatial patterns than those observed in the rental market. A total of **12 High-High clusters** were identified, concentrated in Salzburg's inner urban area and several southeastern towns (towns surrounding Wolfgangsee). This suggests that **sale prices form multiple high-value clusters**, not limited to the urban core, also attracted by beautiful scenery. Simultaneously, **13 Low-Low clusters** were observed, forming a contiguous **low-price belt** in the southern and southeastern parts of the study area—indicative of systemic undervaluation or limited demand.

Furthermore, **2 High-Low and 1 Low-High clusters** highlight localized spatial outliers, pointing to **transitional or mixed-use zones** that deviate from the surrounding market patterns. These may be locations undergoing development, experiencing housing mismatch, or reflecting policy-driven investment zones. A total of **90 units** were classified as not significant, and **16 units** remained undefined due to insufficient spatial neighbors.

These findings support the presence of **clear spatial structure and polarization in the sale apartment market**, with stronger clustering patterns at both ends of the price spectrum compared to rental markets. The stronger presence of Low-Low clusters indicates a potential for **persistent low-value areas**, which could inform regional planning, housing affordability strategies, or future development priorities.

The spatial lag model for sale apartment prices revealed strong and highly significant spatial dependence ($\rho = 0.522$, $p < 0.001$), indicating that local property prices are substantially influenced by those in surrounding areas. This finding is corroborated by the LISA results, which show a positive global spatial autocorrelation (Moran's I = 0.272, $p = 0.001$) and clear clustering patterns across the region.

High-High clusters identified by LISA align with areas featuring good transport accessibility and faster market turnover, supporting the positive regression coefficients for transport infrastructure and listing duration. Conversely, Low-Low clusters dominate southern and peripheral regions, consistent with the negative impacts of migration imbalance and locational disadvantage identified in the regression model. Additionally, the presence of High-Low and Low-High outliers in the LISA map, along with significant heteroskedasticity (Breusch-Pagan $p = 0.018$), suggests spatial heterogeneity that is not fully captured by the global model. These patterns point to localized housing dynamics and call for further exploration using geographically weighted regression.

Discussion

Spatial regression analysis

Rental prices are more sensitive to demographic and spatial structure (age, density, distance to centers). Sales prices are more affected by market dynamics (listing time, size, transaction flow), highlighting speculative or investment-driven behavior. The detailed differences among variable performances are as follows:

Variable	Rental Apartment	Sales Apartment
AvDday	Not significant	-14.00, $p < 0.001$ (significant)
AvM2	Not significant	-6.36, $p = 0.051$ (marginal)
G65_Rat	+0.395, $p < 0.001$ (significant)	+62.90, $p = 0.103$ (not significant)
Emp_Den	+0.0225, $p = 0.015$ (significant)	Not significant
NEU_rat	+0.143, $p = 0.10$ (marginal)	Not significant
Pop_Densit	-0.0117, $p = 0.012$ (significant)	Not significant
Transport	Not significant	+8.44, $p = 0.083$ (marginal)
NEAR_DIST	-0.00011, $p = 0.025$ (significant)	-0.025, $p = 0.129$ (not significant)
F_in_out__	Not significant	-151.83, $p = 0.018$ (significant)

Both markets are shaped by spatial processes, but sales prices exhibit stronger spatial autocorrelation and are more influenced by market behavior, while rental prices are more responsive to demographic and structural spatial features. The contrast highlights the different mechanisms governing rental versus ownership housing markets in spatial terms.

Spatial autocorrelation analysis

A comparison of the LISA cluster maps for rental and sale apartment prices reveals distinct spatial patterns in Salzburg's housing market. While both markets exhibit High–High clusters concentrated around the urban core, the sale market shows more extensive clustering, including peripheral urbanized areas. Notably, Low–Low clusters are significantly more prevalent in the sale market (13 vs. 5), suggesting a stronger polarization effect in property

ownership prices compared to rental prices. Conversely, the rental market displays slightly more spatial outliers (5 vs. 3), indicating greater local variability possibly tied to short-term demand factors.

Feature	Rental Apartments (RA)	Sale Apartments (SA)	Observation
High–High Clusters	11	12	Slightly more in the sale market; indicates broader high-value clustering
Low–Low Clusters	5	13	Sale market shows significantly more low-value clustering → stronger spatial polarization
High–Low & Low–High Outliers	5 (3 High-Low + 2 Low-High)	3 (2 High-Low + 1 Low-High)	Rental market has more spatial outliers, suggesting more local variation
Hotspot Spatial Distribution	Concentrated in Salzburg city core	More dispersed, extends to southeastern towns surrounding Wolfgangsee	Sale market shows a more polycentric structure
Coldspot Spatial Distribution	Scattered, mainly in southern periphery	Forms a continuous belt of low-value areas	Sale market has stronger spatial polarization
Undefined Units	35	16	Sale apartment analysis covers a larger portion of the study area (fewer missing neighbors)

These differences reflect the distinct spatial logic underlying the two markets. The rental market appears more sensitive to employment and demographic dynamics, while the ownership market responds more to structural factors such as accessibility, speculation, and long-term investment trends. The spatial misalignment between rental and sale price clusters also highlights areas of potential housing mismatch, warranting further investigation into tenure dynamics and affordability.

CONCLUSION

The comprehensive analysis of the housing market in Salzburg and its surrounding regions has revealed a complex set of interrelated factors shaping current dynamics. Salzburg has emerged as the most expensive rental region in Austria. This is primarily due to a combination of restricted land availability, high construction and land costs, a growing population, and sustained demand, particularly in the city center.

The study shows that small rental apartments, especially in Salzburg's urban core and commuter suburbs—command disproportionately high rents per square meter. This is largely a result of limited supply and high demand, particularly from students, young professionals, and low-income households. The rental market is highly competitive, especially for entry-level housing.

In contrast, the sales market exhibits a different pattern. Larger apartments tend to achieve higher price per square meter values, while smaller units often show lower values, despite their strong rental demand. This discrepancy is likely due to a combination of older housing stock in the small-unit segment, lower appeal to owner-occupiers, and the inflationary influence of the luxury segment in larger units.

Spatial analyses (including spatial lag models and LISA cluster diagnostics) confirm the existence of strong spatial polarization. High-value clusters are concentrated in central and touristic areas (like Wolfgangsee), while low-value zones dominate rural peripheries. Transitional zones suggest localized dynamics such as gentrification, development pressure, or speculative investment.

Short-term rentals, particularly via platforms like Airbnb, have significantly impacted the housing market in Salzburg. Central urban neighborhoods are most affected, with measurable withdrawal of long-term rental stock. These effects are exacerbated by the profit potential of short-term rentals, which can yield significantly higher returns than conventional leasing.

Moreover, the report provides strong evidence for the financialization of housing in Salzburg. Housing is increasingly treated as an investment asset, rather than as a social good. Investor behavior, limited public housing options, and restrictive zoning (e.g. green space declarations and UNESCO-related height restrictions) further amplify the affordability crisis and contribute to urban displacement and commuting burdens.

Based on the results of this study, the main research questions can be addressed as follows:

1. *What factors have contributed to Salzburg becoming the most expensive rental region in Austria?*

Several interlocking factors have contributed to Salzburg's position as Austria's most expensive rental market. First, the city is geographically constrained, with large portions of land designated as protected green space, which severely limits development potential. Second, Salzburg's status as a UNESCO World Heritage site imposes strict regulations on

building heights and urban transformation. Third, the strong tourism industry creates seasonal and year-round demand for housing, both for short-term visitors and service sector employees. In addition, the city has historically lacked an active land policy and holds very limited public land reserves, which restricts its capacity to shape housing supply. Finally, the shortage of affordable and subsidized housing options exacerbates demand-side pressure in the private rental market, pushing prices upward.

2. What are the factors influencing the dynamics of the housing market in Salzburg?

The dynamics of the Salzburg housing market are shaped by a wide range of factors. On the macro level, interest rate fluctuations and inflation have influenced purchasing power and investment behavior. Regulatory frameworks, such as mandatory parking space requirements and zoning restrictions, directly affect construction costs and supply flexibility. Demographic developments, including an aging population and migration flows, contribute to demand heterogeneity. Moreover, the rise of short-term rentals and investment-oriented purchases have intensified competition for central housing units. While rental prices are most affected by spatial structure and population characteristics, sales prices appear to be more responsive to market behavior, such as listing duration and location-based accessibility.

3. What are the spatial patterns and segmentation within the Salzburg housing market?

The spatial analysis reveals a clear segmentation of the housing market. High-price clusters are predominantly found in central Salzburg and selected touristic locations, such as those near lakes or scenic alpine areas. These areas are characterized by high accessibility, cultural capital, and investor interest. In contrast, low-price clusters are concentrated in peripheral rural areas with weaker infrastructure and fewer economic opportunities. Furthermore, transitional areas show characteristics of spatial heterogeneity, with signs of emerging gentrification or latent development potential. The existence of such spatial patterns confirms the presence of socio-spatial inequality in the Salzburg housing market.

4. How has financialization influenced investment behavior and housing development?

Financialization has had a profound impact on Salzburg's housing market. Housing is increasingly commodified and treated as a financial asset. This is particularly evident in the growth of short-term rental platforms, which offer significantly higher returns than long-term leasing and incentivize landlords to withdraw units from the traditional rental market. Moreover, investor interest in high-yield urban properties has pushed prices upward, making housing less accessible for local residents. Simultaneously, constraints on new housing development within the city—due to land use policies and historical preservation laws—have shifted construction activity to tourist-dominated surrounding regions. These trends reinforce patterns of displacement, longer commutes, and an erosion of housing's social function.

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APPENDIX

Appendix A:

SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION

Data set: merge0721

Spatial Weight: queen=1

Dependent Variable: AvPriM2_RA Number of Observations: 99

Mean dependent var: 14.9859 Number of Variables: 11

S.D. dependent var: 3.32994 Degrees of Freedom: 88

Lag coeff. (Rho): 0.300275

R-squared: 0.415389 Log likelihood: -234.367

Sq. Correlation: - Akaike info criterion: 490.734

Sigma-square: 6.48246 Schwarz criterion: 519.28

S.E of regression: 2.54607

Variable	Coefficient	Std.Error	z-value	Probability
W_AvPriM2_RA	0.300275	0.0807488	3.71863	0.00020
CONSTANT	3.24935	2.78479	1.16682	0.24328
AvDDay_RA	-8.82081e-05	0.00302259	-0.029183	0.97672
AvM2_RA	-0.00843444	0.0122793	-0.686885	0.49215
Pop_Densit	-0.0116682	0.00462864	-2.52087	0.01171
G65_Rat	0.395257	0.10355	3.81707	0.00014
Emp_Den	0.0225138	0.00926555	2.42984	0.01511
NEU_rat	0.14305	0.0868999	1.64615	0.09973
F_in_out__	-0.165385	0.16793	-0.984843	0.32470
Transport	0.0133883	0.0134982	0.99186	0.32127
NEAR_DIST	-0.000109066	4.8512e-05	-2.24823	0.02456

REGRESSION DIAGNOSTICS

DIAGNOSTICS FOR HETEROSKEDASTICITY

RANDOM COEFFICIENTS

TEST	DF	VALUE	PROB
Breusch-Pagan test	9	9.3753	0.40337

DIAGNOSTICS FOR SPATIAL DEPENDENCE

SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : queen=1

TEST	DF	VALUE	PROB
Likelihood Ratio Test	1	11.7932	0.00059

Appendix B

SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION

Data set: merge0721

Spatial Weight: queen=1

Dependent Variable: AvPriM2_SA Number of Observations: 118

Mean dependent var: 4702.88 Number of Variables: 11

S.D. dependent var: 1464.43 Degrees of Freedom: 107

Lag coeff. (Rho): 0.522342

R-squared: 0.502691 Log likelihood: -991.359

Sq. Correlation: - Akaike info criterion: 2004.72

Sigma-square: 1.06651e+06 Schwarz criterion: 2035.2

S.E of regression: 1032.72

Variable	Coefficient	Std.Error	z-value	Probability
W_AvPriM2_SA	0.522342	0.0813494	6.42097	0.00000
CONSTANT	2508.48	920.798	2.72424	0.00644
G65_Rat	62.897	38.5524	1.63147	0.10279
Transport	8.44277	4.86427	1.73567	0.08262
AvDday_SA	-14.0001	2.16821	-6.45699	0.00000
AvM2_SA	-6.35979	3.25831	-1.95187	0.05095
Emp_Den	0.59989	3.48625	0.172073	0.86338
NEU_rat	17.7032	32.7677	0.540264	0.58902
NEAR_DIST	-0.0247263	0.0162911	-1.51778	0.12907
Pop_Densit	-0.374267	1.77148	-0.211273	0.83267
F_in_out__	-151.829	63.941	-2.37451	0.01757

REGRESSION DIAGNOSTICS

DIAGNOSTICS FOR HETEROSKEDASTICITY

RANDOM COEFFICIENTS

TEST	DF	VALUE	PROB
Breusch-Pagan test	9	20.0122	0.01784

DIAGNOSTICS FOR SPATIAL DEPENDENCE

SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX: queen=1

TEST	DF	VALUE	PROB
Likelihood Ratio Test	1	26.6490	0.00000